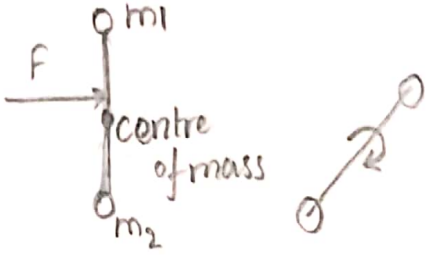
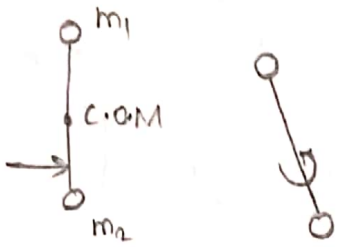
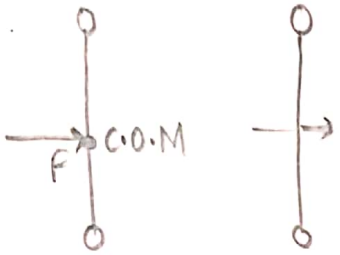
* Centre of Mass





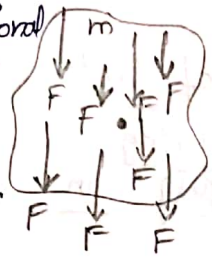


* CM → entire mass of body appears to be concentrated.
* CM is used to describe the motion of the body.

* It may be inside or outside the body but within the boundaries of body.
* There is no need of mass present at the centre.

Centre of gravity:

* C.G → resultant gravitational force (weight) of the body appears to be concentrated.

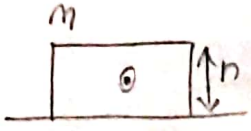


* It must lie inside the body.
* It is used to explain the stability of body.
* In uniform gravity & for symmetric shaped bodies — COM & COG do coincide.



* $COG = \frac{\int m_i g x_i}{\int m_i g}$ ⇒ If g = const then $C.G = \frac{\int m_i x_i}{\int m_i}$

*



$U = \Sigma m_i g h_i = g \Sigma m_i h_i$

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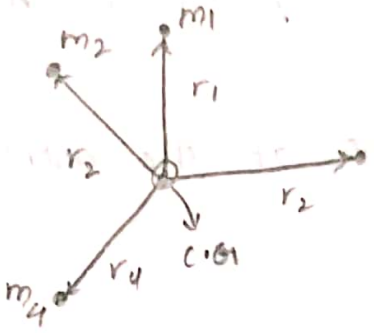
$$U = g \sum m_i \frac{\sum m_i h_i}{\sum m_i}$$

$$\boxed{U = Mgh/2}$$

Moment of mass:

* 

mass moment $= m\bar{r}$.

* 

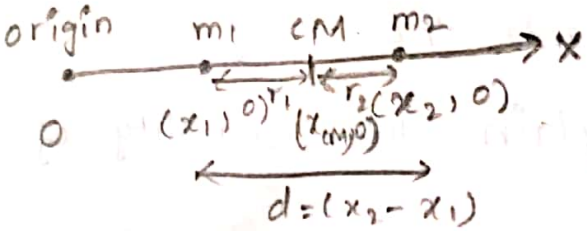
About C.G, $\boxed{\sum m_i \bar{r_i} = 0}$

Algebraic sum of mass moment about C.G is zero.

i.e, $m_1\bar{r_1} + m_2\bar{r_2} + \ldots = 0$.

* Co-ordinates of Centre of Mass:

① Particles lying on x-axis:-



About C.M, $\sum m_i r_i = 0$

~~$(-m_1 r_1) + m_2 r_2$~~

$m_1 r_1 (-\uparrow) + m_2 r_2 (\uparrow) = 0$

$$\boxed{m_1 r_1 = m_2 r_2}$$

i.e, if $m_1 > m_2$ then $r_1 < r_2$.

if $m_1 < m_2$ then $r_1 > r_2$.

i.e, CM is closer to heavier

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* from figure,

$r_1 = x_{CM} - x_1$

$r_2 = x_{CM} + x_2$

$\therefore \quad m_1(x_{CM} - x_1) = m(x_2 - x_{CM})$

$m_1 x_{CM} - m_1 x_1 = m_2 x_2 - m_2 x_{CM}$

$(m_1 + m_2) x_{CM} = m_1 x_1 + m_2 x_2$

$$\boxed{x_{CM} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}}$$

Again,

$m_1 r_1 = m_2 r_2 \qquad [r_1 + r_2 = d]$

$m_1 r_1 = m_2 (d - r_1)$ & $m_1(d - r_2) = m_2 r_2$

$m_1 r_1 + m_2 r_1 = m_2 d$ $\qquad m_1 d = (m_1 + m_2) r_2$

$d = \frac{(m_1 + m_2) r_1}{m_2}$

$$* \boxed{r_2 = \frac{m_1 d}{m_1 + m_2}} *$$

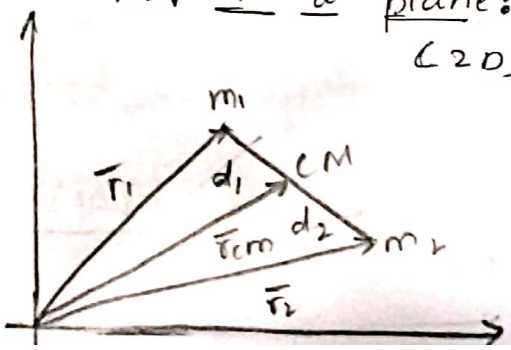
$$* \boxed{r_1 = \frac{m_2 d}{m_1 + m_2}} *$$

* If n-particles are lying along y-axis,

$$y_{cm} = \frac{m_1 y_1 + m_2 y_2 + \cdots + m_n y_n}{m_1 + m_2 + \cdots + m_n}$$

& $$z_{cm} = \frac{m_1 z_1 + m_2 z_2 + \cdots + m_n z_n}{m_1 + m_2 + \cdots + m_n}$$

* Particles lying in a plane:



(20) $\bar{r}_{cm} + \bar{d}_1 = \bar{r}_1$

$\Rightarrow \bar{d}_1 = \bar{r}_1 - \bar{r}_{cm}$

& $\bar{r}_{cm} + \bar{d}_2 = \bar{r}_2$

$\bar{d}_2 = \bar{r}_2 - \bar{r}_{cm}$

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$$m_1 \bar{d}_1 + m_2 \bar{d}_2 = 0$$

$$m_1(\bar{r}_1 - \bar{r}_{cm}) + m_2(\bar{r}_2 - \bar{r}_{cm}) = 0$$

* $$\boxed{\frac{m_1\bar{r}_1 + m_2\bar{r}_2}{m_1 + m_2} = \bar{r}_{cm}}$$

For n-particles,

$$\bar{r}_{cm} = \frac{m_1\bar{r}_1 + m_2\bar{r}_2 + \cdots + m_n\bar{r}_n}{m_1 + m_2 + \cdots + m_n}$$

* $\bar{r}_1 = x_1\hat{i} + y_1\hat{j}$

$\bar{r}_2 = x_2\hat{i} + y_2\hat{j}$

$\Rightarrow \bar{r}_{cm} = x_{cm}\hat{i} + y_{cm}\hat{j}$

* $$x_{cm}\hat{i} + y_{cm}\hat{j} = \frac{m_1(x_1\hat{i} + y_1\hat{j}) + m_2(x_2\hat{i} + y_2\hat{j})}{m_1 + m_2}$$

comparing $\hat{i}$, $\hat{j}$ co-ordinates,

$$x_{cm} = \frac{m_1x_1 + m_2x_2}{m_1 + m_2}, \quad y_{cm} = \frac{m_1y_1 + m_2y_2}{m_1 + m_2}$$

* co-ordinates of CM = (x_{cm}, y_{cm})

$$d_{cm} = \sqrt{x_{cm}^2 + y_{cm}^2}$$

Particles lying in 3D:

* co-ordinates of CM = (x_{cm}, y_{cm}, z_{cm})

* distance of CM from origin,

$$d = \sqrt{x_{cm}^2 + y_{cm}^2 + z_{cm}^2}$$

* Points:-

* $$x_{cm} = \frac{m_1x_1 + m_2x_2}{m_1 + m_2}$$

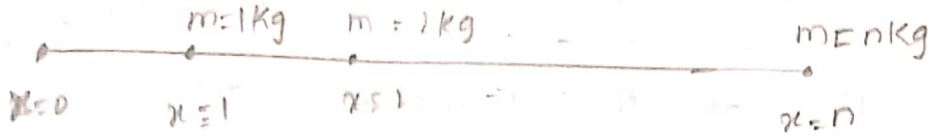
displacement of CM,

$$\Delta x_{cm} = \frac{m_1\Delta x_1 + m_2\Delta x_2}{m_1 + m_2}$$

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* If $\Delta x_{cm} = 0 \Rightarrow \boxed{m_1 \Delta x_1 + m_2 \Delta x_2 = 0}$

Eg:

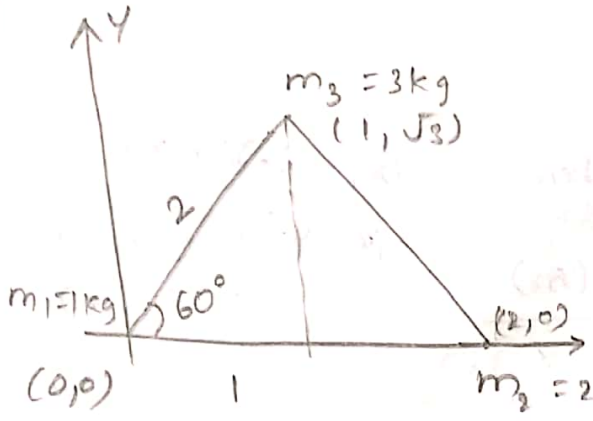


$$x_{cm} = \frac{1(1) + 2(2) + \dots + n(n)}{1+2+\dots+n}$$

$$= \frac{\cancel{n(n+1)}(2n+1)}{\cancel{6}\ 3} \times \frac{\cancel{6}}{\cancel{n(n+1)}}$$

$$= \frac{2n+1}{3}.$$

→ Eg:

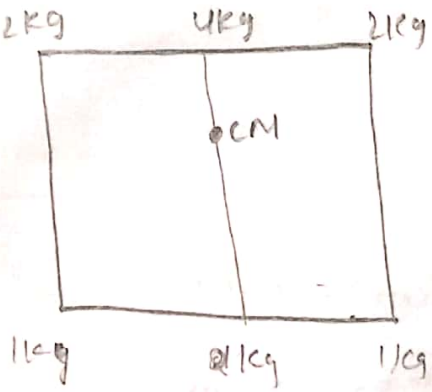


$$x_{CM} = \frac{1\times0 + 2(2) + 3(1)}{1+2+3}$$

$$= \frac{4+3}{6} = \frac{7}{6}$$

$$y_{cm} = \frac{1\times0 + 0\times2 + 3\times\sqrt{3}}{6} = \frac{3\sqrt{3}}{6} = \frac{\sqrt{3}}{2}$$

→



From figure,

$$r_{\text{from } 2kg} = \frac{4\times4}{4+2} = \frac{16}{6} = \frac{8}{3} \text{ cm}$$

Methods used to locate the position of C.M:-

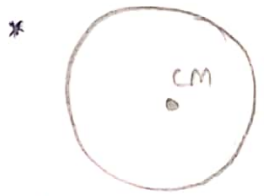
1) Method of symmetry 2) Method of decomposition.

3) Integration method.

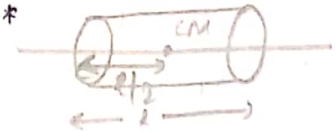
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① Method of Symmetry :-

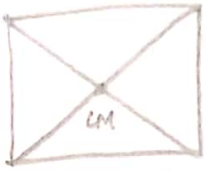
* For uniform mass distribution & symmetry shaped bodies, C.M lies at the geometrical centre.



ring/disc/sphere



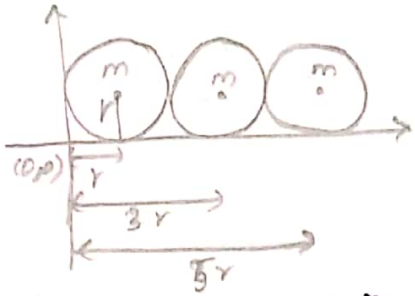
cylinder



square plate



rectangular plate



All are identical spheres.

$(x_1, y_1) = (r, r)$

$(x_2, y_2) = (3r, r)$

$(x_3, y_3) = (5r, r)$

$$x_{cm} = \frac{m(r) + m(3r) + m(5r)}{3m}$$

$$= \frac{9r}{3} = 3r$$

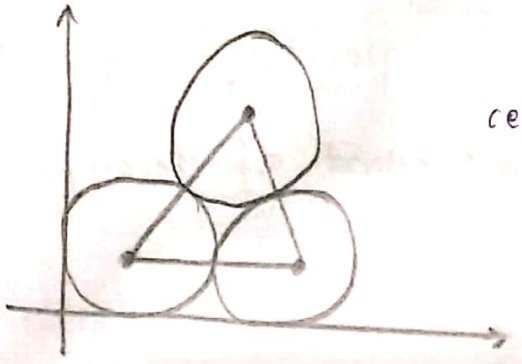
$$y_{cm} = \frac{m(r) + m(r) + m(r)}{3m}$$

$$= 3r/3 = r.$$

co-ordinates of C.M $= (x_{cm}, y_{cm}) = (3r, r)$

Distance of CM from origin $= \sqrt{x_{cm}^2 + y_{cm}^2}$

*

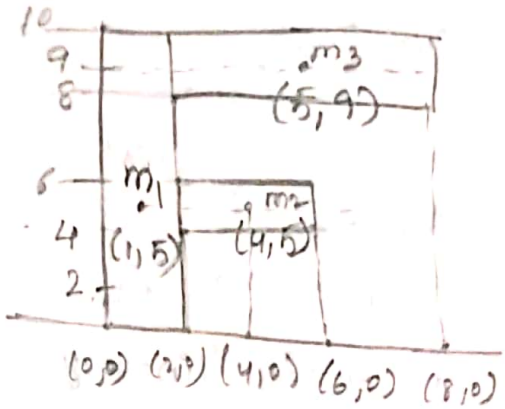


centre of mass will be at the point joining the medians of the triangle i.e, centroid of the equilateral Δle.

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Method of decomposition :-

σ = mass per unit area = const



$m_1 = \sigma A_1$

$= \sigma (2 \times 10)$

$= 20\sigma$

$m_2 = \sigma A_2 = \sigma (4 \times 2) = 8\sigma$

$m_3 = \sigma A_3 = \sigma (6 \times 2) = 12\sigma$

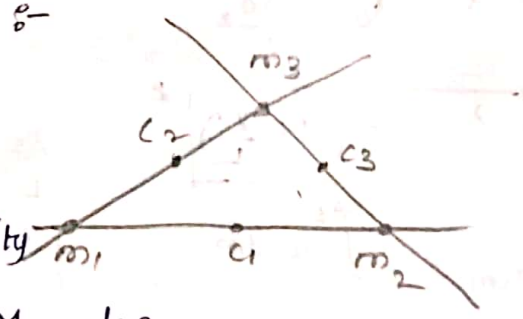
$$x_{CM} = \frac{20\sigma(1) + 8\sigma(4) + 12\sigma(5)}{40\sigma}$$

$$= \frac{20 + 32 + 60}{40} = \frac{122}{40} = \frac{61}{20}.$$

$$y_{cm} = \frac{20\sigma(5) + 8\sigma(5) + 12\sigma(9)}{40\sigma}$$

$$= \frac{100 + 40 + 108}{40} = \frac{248}{40} = \frac{62}{10}$$

Method of integration :-

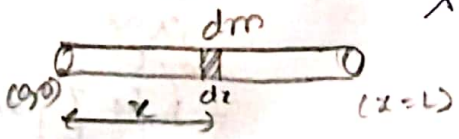


a) For thin rod :

① λ = linear mass density

= const

$$\lambda = \frac{M}{L} = \frac{dm}{dx}$$



* For a discrete particle system,

$$x_{cm} = \frac{\Sigma m_i x_i}{\Sigma m_i}.$$

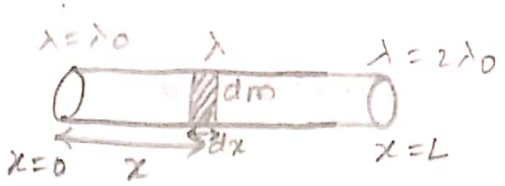
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For continuous mass distribution,

$$x_{cm} = \frac{\int_{x=0}^{L} (dm)\, x}{\int_0^L dm} = \frac{\int_0^L (\lambda \cdot dx) \cdot x}{\int_0^L \lambda \cdot dx}$$

$$x_{cm} = \frac{\lambda \left[\frac{x^2}{2}\right]_0^L}{\lambda [x]_0^L} = \frac{L^2}{2(L)} = \frac{L}{2}.$$

(Q) If λ varies linearly from λ_0 to $2\lambda_0$ for a length L.



we have,

$$\lambda = a + bx$$

when $x = 0$, $\lambda = \lambda_0$

$$\boxed{\lambda_0 = a}$$

when $x = L$, $\lambda = 2\lambda_0$

$$2\lambda_0 = a + bL$$

$$2\lambda_0 = \lambda_0 + bL$$

$$\boxed{\frac{\lambda_0}{L} = b}$$

$$\Rightarrow \lambda = \lambda_0 + \frac{\lambda_0}{L} x$$

$$\boxed{\lambda = \lambda_0 \left(1 + \frac{x}{L}\right)} \quad *$$

$$x_{cm} = \frac{\int_0^L x \cdot dm}{\int_0^L dm} = \frac{\int_0^L x \cdot \lambda \cdot dx}{M}$$

$$= \frac{\int_0^L x \cdot \lambda_0 \left(1 + \frac{x}{L}\right) \cdot dx}{M}$$

$$= \frac{\lambda_0}{M} \int_0^L \left(x + \frac{x^2}{L}\right) \cdot dx$$

$$M = \int_0^L dm$$

$$= \int_0^L x \cdot dx$$

$$= \int_0^L \lambda_0 \left(1 + \frac{x}{L}\right) \cdot dx$$

$$\boxed{M = \frac{3L\lambda_0}{2}} = \lambda_0 \left(L + \frac{1}{L} \frac{L^2}{2}\right)$$

$$= \lambda_0 \left(L + \frac{L}{2}\right)$$

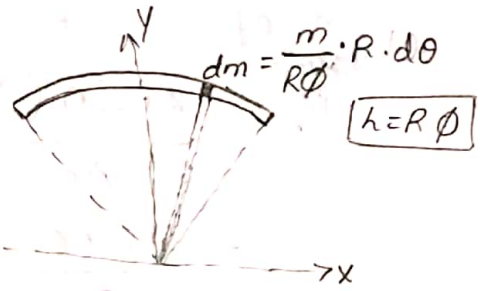
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$$= \frac{\lambda_0}{M}\left[\frac{L^2}{4} + \frac{3L^2}{4}\right] \quad \frac{\lambda_0}{M}\left[\frac{L^2}{2} + \frac{2L^2}{3\times 2}\right]$$

$$= \frac{\lambda_0}{M}\left[\frac{5L^2}{6}\right] = \frac{2\lambda_0}{3L}\left[\frac{5L^2}{6_3}\right] = \frac{5L}{9}.$$

Centre of mass of circular arc shaped ring/wire:

$$y_{cm} = \frac{\int dm \cdot y}{\int dm} = \frac{-\int_{-\phi/2}^{\phi/2} \frac{m}{\phi} \cdot d\theta \cdot R\cos\theta}{m}$$



$$= \frac{m}{\phi} R \left[\sin\theta\right]_{-\phi/2}^{\phi/2}$$

$$= \frac{R}{\phi}\left[\sin\phi/2 + \sin\phi/2\right]$$

$$\boxed{y_{cm} = \frac{2R\sin\phi/2}{\phi}}$$

* for quarter circle:

$$r_{cm} = \frac{2R\sin(\pi/4)}{\pi/2}$$

$$= \frac{2\sqrt{2}R}{\pi}$$



position co-ordinates from origin

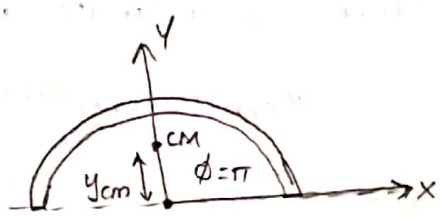
$$x_{com} = r_{cm}\cos 45° = \frac{2R}{\pi}$$

$$y_{com} = r_{cm}\sin 45° = \frac{2R}{\pi}$$

$$(x_{com}, y_{com}) = \left(\frac{2R}{\pi}, \frac{2R}{\pi}\right)$$

* for semi circle:

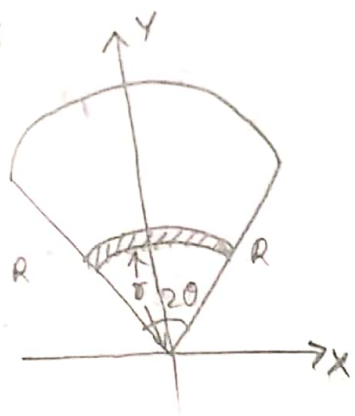
$$y_{cm} = \frac{2R\sin(\pi/2)}{\pi/2}$$



$$\boxed{y_{cm} = \frac{2R}{\pi}}$$

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Centre of mass of arc shaped disc :



mass of element

$= \frac{M}{A} \times$ area of small element (dA)

$$dm = \frac{M}{R^2\pi} \times 2\pi r dr \cdot \theta = \frac{2M}{R^2} r \cdot dr$$

2π rad $\longrightarrow \pi R^2$

1 rad $\longrightarrow \frac{R^2}{2}$

$2\theta \longrightarrow \frac{R^2}{2} \times 2\theta = R^2\theta$

$dA = A_2 - A_1 = (r+dr)^2\theta - r^2\theta$

$= 2rdr\theta$

COM of element,

$$r_{cm} = \frac{2r\sin\left(\frac{2\theta}{2}\right)}{2\theta} = \frac{r\sin\theta}{\theta}$$

$$y_{cm} = \frac{\int dm \cdot r_{cm}}{\int dm} = \frac{\int_{r=0}^{r=R} \frac{2M}{R^2} \cdot r dr \cdot \frac{r\sin\theta}{\theta}}{M}$$

$$= \frac{2M}{R^2} \frac{\sin\theta}{\theta} \left[\frac{r^3}{3}\right]_0^R \times \frac{1}{M}$$

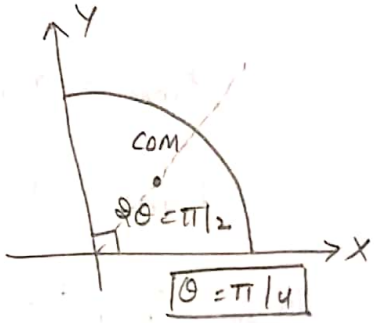
$$= \frac{2}{R^2} \times \frac{\sin\theta}{\theta} \cdot \frac{R^3}{3}$$

$$\boxed{y_{cm} = \frac{2R\sin\theta}{3\theta}}$$

* for quater circular disc :

$$r_{cm} = \frac{2R\sin(\pi/4)}{3(\pi/4)}$$

$$r_{com} = \frac{4R\sqrt{2}}{3\pi}$$



∴ position co-ordinates of COM

$$x_{cm} = r_{cm}\cos 45^\circ$$

$$y_{com} = r_{cm}\sin 45^\circ$$

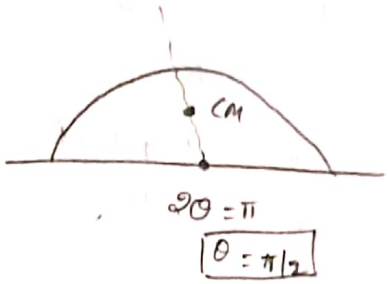
$$(x_{cm}, y_{cm}) = \left(\frac{4R}{3\pi}, \frac{4R}{3\pi}\right)$$

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* for semi circular disc:

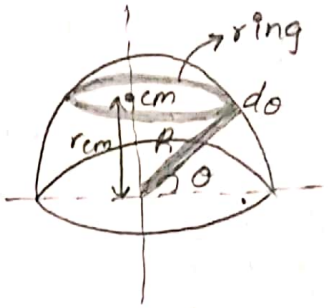
$$y_{cm} = \frac{2R\sin(\pi/2)}{S(\pi/2)}$$

$$= \frac{4R}{3\pi}$$



Centre of mass of hemispherical shell (Hollow sphere):

$\cos\theta = \frac{r}{R} \Rightarrow r = R\cos\theta$



$r_{cm} = R\sin\theta$

$dm = \frac{m}{A} \cdot dA$

$= \frac{m}{2\pi R^2} \times 2\pi r \cdot d\theta \cdot R$

$= \frac{M}{R} \times R\cos\theta \cdot d\theta$

$= M\cos\theta \cdot d\theta$

$$y_{cm} = \frac{\int r_{cm} \cdot dm}{\int dm} = \frac{\int R\sin\theta \cdot M\cos\theta \cdot d\theta}{M}$$

$$= \frac{R}{2} \int_{\theta=0}^{\theta=\pi/2} \sin 2\theta \cdot d\theta$$

$$= \frac{R}{2} \left[-\frac{\cos 2\theta}{2} \right]_0^{\pi/2}$$

$$= \frac{R}{4} [2] = R/2.$$

Centre of mass of solid hemisphere:

$dm = \frac{M}{V} \cdot dV$

$= \frac{M}{\frac{2}{3}\pi R^3} \cdot 2\pi r^2 \cdot dr$

$= \frac{3Mr^2 \cdot dr}{R^3}$

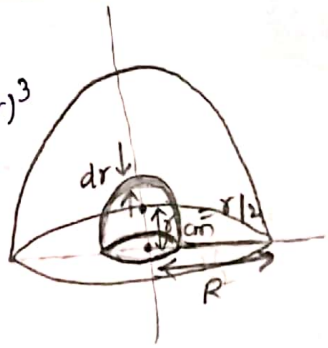
$y_{cm} = \frac{\int r_{cm} \cdot dm}{\int dm}$

$dV = V_2 - V_1$

$= \frac{2}{3}\pi(r+dr)^3 - \frac{2}{3}\pi r^3$

$= \frac{2}{3}\pi(3r^2 \cdot dr)$

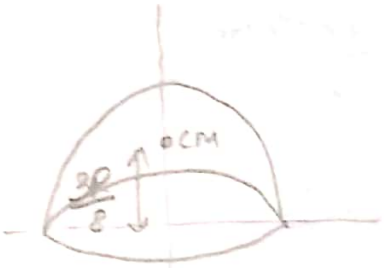
$= 2\pi r^2 \cdot dr$



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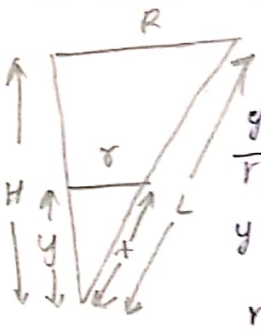
$$= \frac{\int_{r=0}^{r=R} \frac{r}{2} \cdot \frac{3Mr^2 \cdot dr}{R^3}}{M}$$

$$= \frac{3}{2R^3} \left[\frac{r^4}{4} \right]_0^R$$



$$\boxed{y_{cm} = \frac{3R}{8}}$$

Centre of Mass of hollow cone :



$$\frac{y}{r} = \frac{H}{R}$$

$$y = \frac{Hr}{R}$$

$$\frac{r}{x} = \frac{R}{L}$$

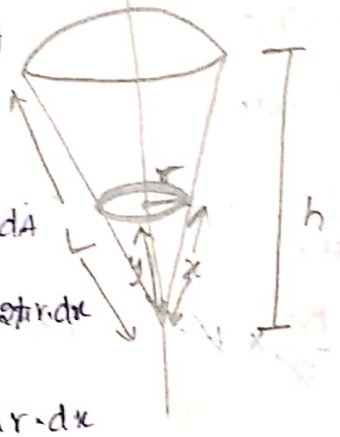
$$r = \frac{Rx}{L}$$

slant ~~height~~ length = L

$$dm = \frac{M}{A} \times dA$$

$$= \frac{M}{\pi RL} \times 2\pi r \cdot dx$$

$$= \frac{2Mr \cdot dx}{RL}$$



$$y_{cm} = \frac{\int y\, dm}{\int dm} = \frac{\int \frac{Hr}{R} \cdot \frac{2Mr \cdot dx}{RL}}{M}$$

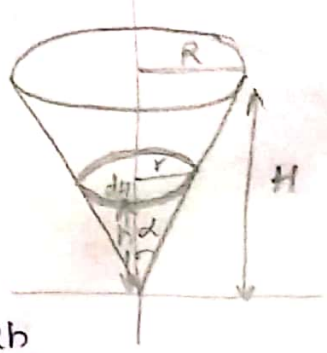
$$= \frac{2H}{R^2L} \int r^2 \cdot dx$$

$$= \frac{2H}{R^2L} \times \frac{R^3}{3} \left[\frac{x^3}{3} \right]_0^L$$

$$\boxed{y_{COM} = \frac{2H}{3}} \rightarrow \text{from apex.}$$

$$\boxed{y_{COM} = \frac{H}{3}} \rightarrow \text{from base}$$

Centre of Mass of solid cone :



$$\frac{R}{H} = \frac{r}{h}$$

$$r = \frac{Rh}{H}$$

$$dm = \frac{M}{V} \cdot dv$$

$$= \frac{M}{\frac{1}{3}\pi R^2 H} \cdot \pi r^2 \cdot dh$$

$$= \frac{3Mr^2 dh}{R^2 H}$$

$$y_{cm} = \frac{\int h \cdot dm}{\int dm}$$

$$= \frac{\int \frac{3Mr^2 \cdot dh}{R^2 H} \cdot h}{M}$$

$$= \frac{3}{R^2 H} \int_0^H \frac{R^2 h^2}{H^2} h \cdot dh$$

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$$= \frac{3}{R^2 H} \times \frac{R^2}{H^2} \left[\frac{h^4}{4}\right]_0^H$$

$$= \frac{3}{H^3} \left(\frac{H^4}{4}\right)$$

$$\boxed{y_{cm} = \frac{3H}{4}}$$

↳ from apex

$$y_{cm} = H - \frac{3H}{4}$$

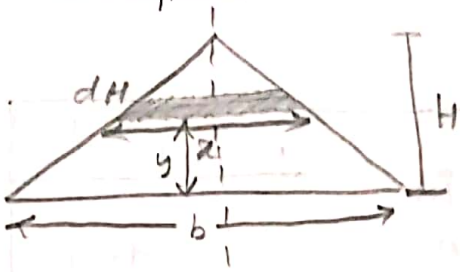
$$\boxed{y_{cm} = \frac{H}{4}}$$

↳ from base

Centre of Mass of triangular plate:

$$dm = \frac{M}{A} \cdot dA$$

$$= \frac{M}{\frac{1}{2} bH} \times dy$$



$$\frac{b}{H} = \frac{x}{H-y}$$

$$dm = \frac{2M}{bH} x \, dy$$

$$y_{cm} = \frac{\int y \cdot dm}{\int dm} = \frac{\int y \cdot \frac{2M}{bH} x \cdot dy}{M} = \frac{2}{bH} \int yx \cdot dy$$

$$\left[\because x = \frac{b(H-y)}{H}\right]$$

$$y_{cm} = \frac{2}{bH} \int_0^H \frac{b(H-y)}{H} \cdot y \, dy$$

$$= \frac{2}{H^2} \left[\int_0^H Hy \cdot dy - \int_0^H y^2 \cdot dy\right]$$

$$= \frac{2}{H^2} \cdot H \left[\frac{y^2}{2}\right]_0^H - \frac{2}{H^2} \left[\frac{y^3}{3}\right]_0^H$$

$$= \frac{H^2}{H} - \frac{2}{H^2} \cdot \frac{H^3}{3}$$

$$= H - \frac{2H}{3}$$

$$\boxed{y_{com} = H/3}$$



$$H = \frac{\sqrt{3}L}{2}$$

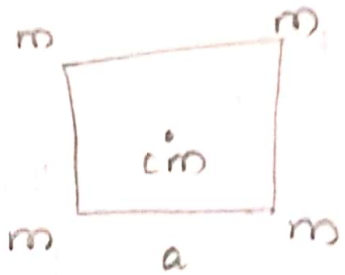
$$y_{com} = \frac{H}{3} = \frac{L}{2\sqrt{3}}$$

Centre of Mass of bodies with cavity:

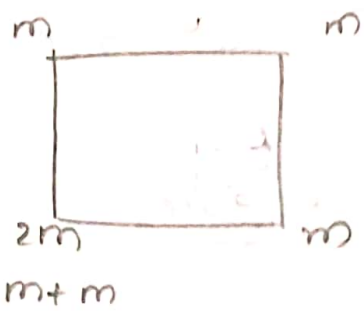
shift in centre of mass is calculated.

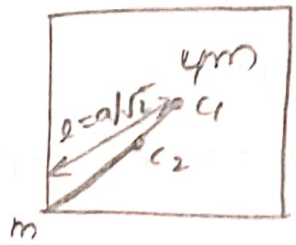
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Particle system



If mass of one of the particle is doubled





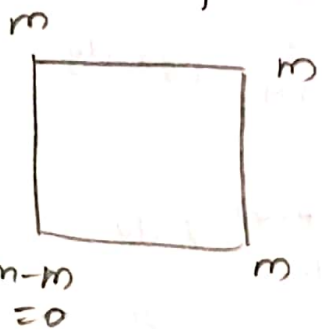
C_1C_2 = shift in cm

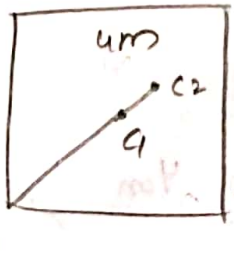
$= \frac{m(a/\sqrt{2})}{5m}$

$= \frac{a}{5\sqrt{2}}$

$$\text{shift in cm} = \frac{m_{added} \times d}{M + m_{added}}$$

If one of the particles is removed.



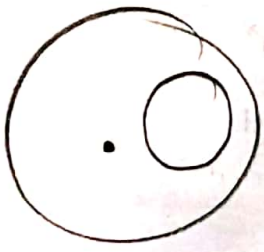


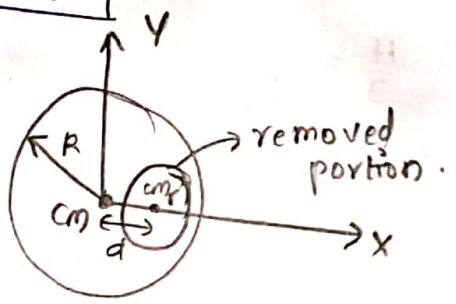
$C_1C_2 = \frac{(-m)a/\sqrt{2}}{4m-m}$

$C_1C_2 = \frac{-a}{8\sqrt{2}}$

$$\text{shift in cm} = \frac{m_{removed}(-ve\ sign) \times d}{m + m_{removed}(-ve\ sign)}$$

Disc:





position of cm of remaining position of disc is

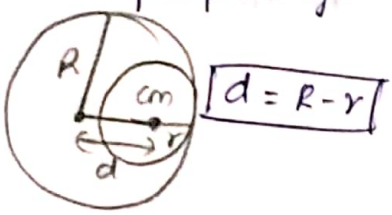
$$\text{Shift in cm} = \frac{m_{removed} \times d}{M_{Total} + m_{removed}}$$

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$$= \frac{-\sigma\pi r^2 d}{\sigma\pi R^2 - \sigma\pi r^2}$$

$$\boxed{x_{shift} = \frac{-r^2 d}{(R^2 - r^2)}}$$

① If mass is removed from periphery.

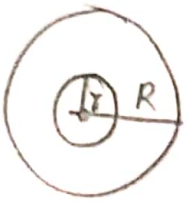


$$\boxed{d = R - r}$$

$$x_{shift} = \frac{-r^2(R-r)}{R^2 - r^2}$$

$$\boxed{x = \frac{-r^2}{R+r}}$$ → shift is maximum

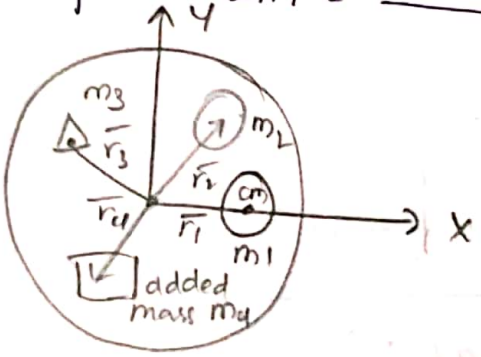
②



$\boxed{d=0}$

$$\boxed{x_{shift} = 0}$$

* shift in com $= \frac{M\bar{r} - m_1\bar{r}_1 - m_2\bar{r}_2 - m_3\bar{r}_3 + m_4\bar{r}_4}{M - m_1 - m_2 - m_3 + m_4}$

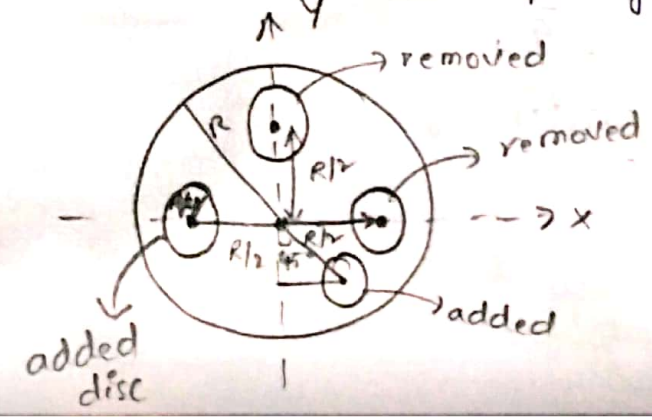


$\boxed{m \propto A}$ → disc, square plate, triangular plate.

for cube (or) sphere $\boxed{m \propto V}$

$\bar{r}$ = position vector of com of original disc.

→

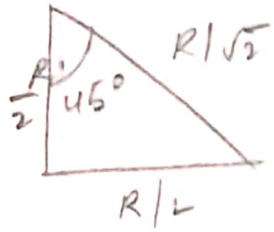


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$m \propto A$

$= \pi \left(\frac{R}{4}\right)^2$

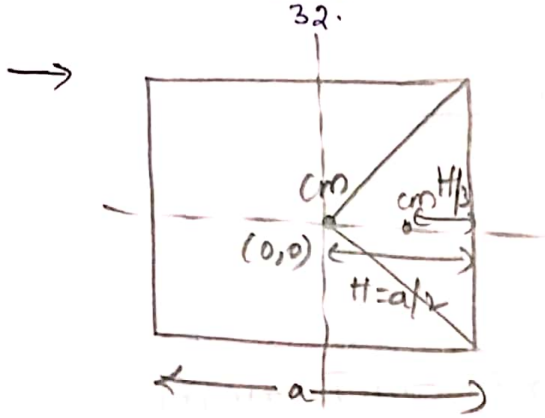
$= \frac{\pi R^2}{16} \sigma$



$M = \sigma \pi R^2$

$\text{shift} = M(0) - \frac{\pi R^2}{16} \times \frac{R}{2}$

$= -\frac{\pi R^3}{32}$



$H = a/2$

$\frac{H}{3} = \frac{a}{6}$

$d = a/6$

$x_{shift} = \frac{M(0) - m\bar{r}_1}{M - m}$

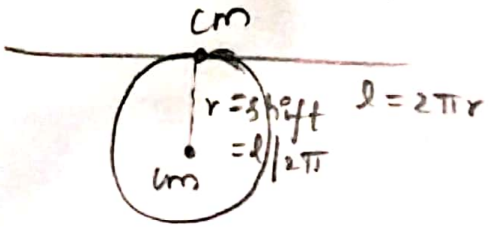
$= \frac{M(0) - \frac{1}{2}(a)\left(\frac{a}{2}\right)\left(\frac{a}{3}\right)}{a^2 - \frac{1}{2}a\left(\frac{a}{2}\right)}$

$= \frac{-\frac{1}{2}\frac{a^3}{6}}{a^2\left(1 - \frac{1}{4}\right)} = \frac{+\frac{a}{12}}{+\frac{3}{4}} = \frac{a}{9}$

$\boxed{x_{shift} = \frac{a}{9}}$

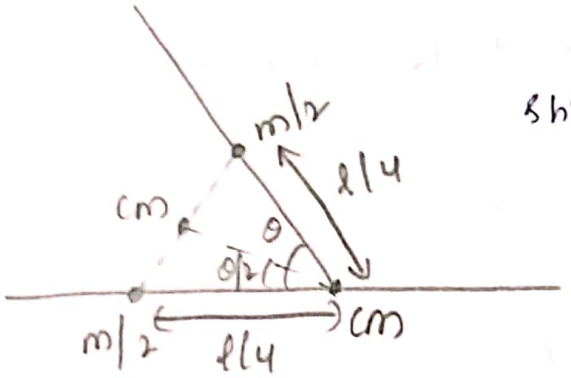
Shift in bending

*



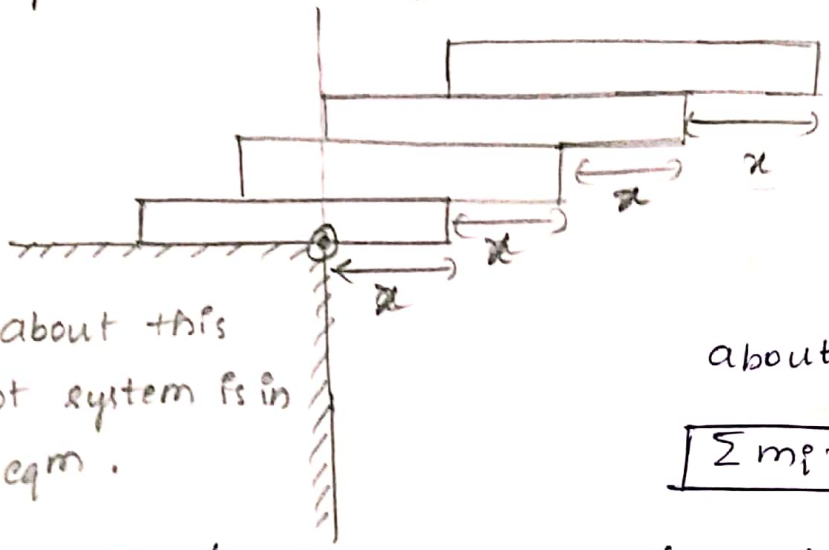
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*



Shift in cm $= \frac{l}{4}\cos\theta/2$

→ Find Max projection of each block so that system is in equilibrium.



about this pt system is in eqm.

i.e, cm of system lies at the edge of table.

about CM,

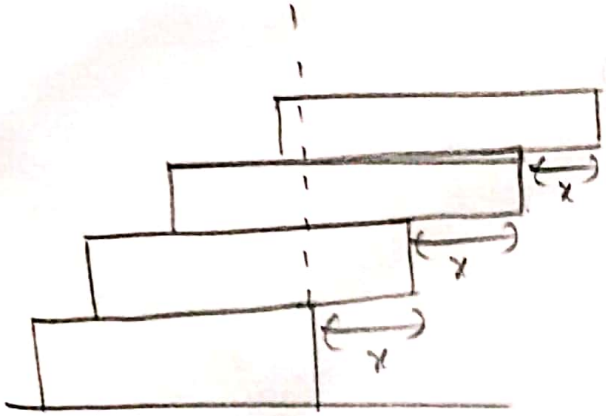
$$\boxed{\Sigma m_i x_i = 0}$$

$$-m(l/2 - x) + m(l/2 - 2x) + m(l/2 - 3x) + m(l/2 - 4x) = 0$$

$$2l - 10x = 0$$

$$\boxed{x = l/5}$$

*



$$\Sigma M_i x_i = 0$$

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$$m(l/2 - x) + m(l/2 - 2x) + (l/2 - 3x)m = 0$$

m

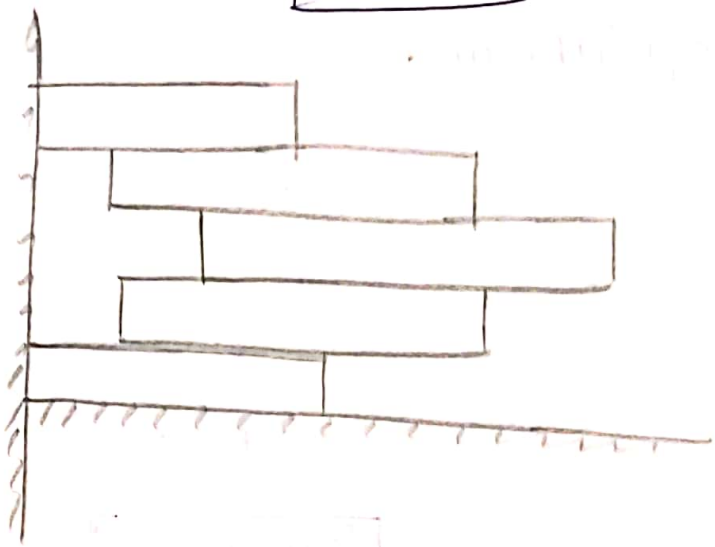
$$\frac{3l}{2} - 6x = 0$$

$$\frac{3l}{2} = 6x$$

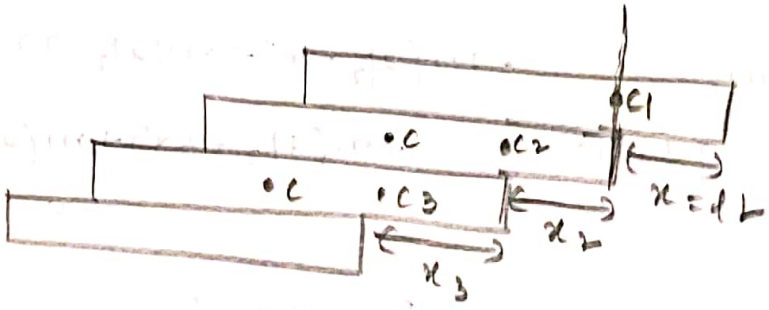
$$\boxed{x = l/4}$$

sl

*



*



$$x_{cm} = \frac{m_1(x_1) + m_2(x_2)}{m_1 + m_2}$$

$$x_2 = \frac{m(0) + m(l/2)}{2m}$$

$$= \frac{ml/2}{2m}$$

$$\boxed{x_2 = l/4}$$

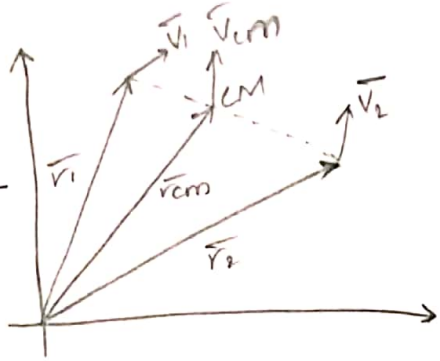
$$x_3 = \frac{2m(0) + m(l/2)}{2m + m} \Rightarrow \boxed{x_3 = l/6}$$

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Motion of Centre of Mass:

Position vector of COM,

$$\vec{r}_{COM} = \frac{m_1\vec{r}_1 + m_2\vec{r}_2 + \cdots + m_n\vec{r}_n}{m_1 + m_2 + \cdots + m_n}$$



displacement of COM,

$$\Delta\vec{r}_{cm} = \frac{m_1\Delta\vec{r}_1 + m_2\Delta\vec{r}_2 + \cdots m_n\Delta\vec{r}_n}{m_1 + m_2 + \cdots + m_n} \quad -(a)$$

Again, on differentiating eqn (a) w.r.to time,

$$\frac{d\vec{r}_{cm}}{dt} = \frac{m_1\frac{d\vec{r}_1}{dt} + m_2\frac{d\vec{r}_2}{dt} + \cdots m_n\frac{d\vec{r}_n}{dt}}{m_1 + m_2 + \cdots + m_n}$$

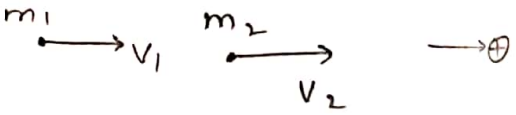
$$\vec{v}_{cm} = \frac{m_1\vec{v}_1 + m_2\vec{v}_2 + \cdots + m_n\vec{v}_n}{m_1 + m_2 + \cdots + m_n} \quad -(b)$$

For two particle system,

$$\boxed{\vec{v}_{cm} = \frac{m_1\vec{v}_1 + m_2\vec{v}_2}{m_1 + m_2}}$$

Points:

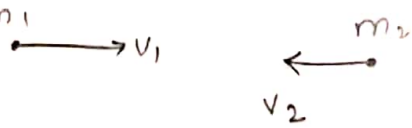
(1) $\vec{v}_{cm} = \dfrac{m_1v_1\hat{i} + m_2v_2\hat{i}}{m_1 + m_2}$



$$\vec{v}_{cm} = \frac{(m_1v_1 + m_2v_2)\hat{i}}{m_1 + m_2}$$

$$\boxed{v_{cm} = \frac{m_1v_1 + m_2v_2}{m_1 + m_2}}$$

(2) $\vec{v}_{cm} = \dfrac{m_1v_1\hat{i} + m_2v_2(-\hat{i})}{m_1 + m_2}$



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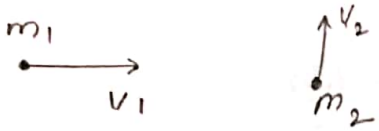
$$\bar{V}_{cm} = \frac{(m_1 v_1 - m_2 v_2)\uparrow}{m_1 + m_2}$$

$$\boxed{V_{cm} = \frac{m_1 v_1 - m_2 v_2}{m_1 + m_2}}$$

If $m_1 v_1 = m_2 v_2$ then $\boxed{V_{com} = 0}$

(3)

$$\bar{V}_{cm} = \frac{m_1 v_1 \uparrow + m_2 v_2 \uparrow}{m_1 + m_2}$$



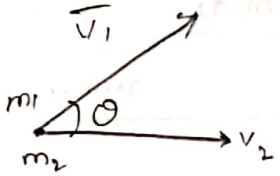
$$\bar{V}_{cm} = \left(\frac{m_1 v_1}{m_1 + m_2}\right)\uparrow + \left(\frac{m_2 v_2}{m_1 + m_2}\right)\uparrow$$

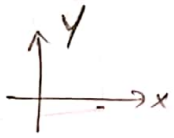
$$\tan\theta = \frac{m_2 v_2}{m_1 v_1}$$



$$\& \quad |\bar{V}_{cm}| = \frac{\sqrt{m_1^2 v_1^2 + m_2^2 v_2^2}}{m_1 + m_2}$$

(4) General formula





$$\bar{V}_{cm} = \frac{m_1(v_1\cos\theta\uparrow + v_1\sin\theta\uparrow) + m_2 v_2 \uparrow}{m_1 + m_2}$$

$$= \frac{(m_1 v_1\cos\theta + m_2 v_2)\uparrow + m_1 v_2 \sin\theta\uparrow}{m_1 + m_2}$$

$$|\bar{V}_{cm}| = \frac{\sqrt{(m_1 v_1\cos\theta + m_2 v_2)^2 + (m_1 v_2 \sin\theta)^2}}{m_1 + m_2}$$

$$|\bar{V}_{cm}| = \frac{\sqrt{m_1^2 v_1^2 + m_2 v_2^2 + 2(m_1 v_1)(m_2 v_2)\cos\theta}}{m_1 + m_2}$$

Again we have,

$$\bar{V}_{cm} = \frac{m_1 \bar{v}_1 + m_2 \bar{v}_2 + \cdots m_n \bar{v}_n}{m_1 + m_2 + \cdots + m_n}$$

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$$(m_1+m_2+---m_n)(\bar{V}_{cm}) = m_1\bar{v}_1 + m_2\bar{v}_2 + \cdots + m_n\bar{v}_n$$

$$\boxed{\bar{P}_{com} = \bar{P}_1 + \bar{P}_2 + \cdots + \bar{P}_n}$$

i.e, net momentum of all particles of system is equal to momentum of COM of system.

Acceleration of COM:

we have,

$$\bar{V}_{cm} = \frac{m_1\bar{v}_1 + m_2\bar{v}_2 + \cdots + m_n\bar{v}_n}{m_1+m_2+\cdots+m_n}$$

on differentiating w.r.to time,

$$\frac{d\bar{V}_{cm}}{dt} = \frac{m_1\frac{d\bar{v}_1}{dt} + m_2\frac{d\bar{v}_2}{dt} + \cdots + m_n\frac{d\bar{v}_n}{dt}}{m_1+m_2+\cdots+m_n}$$

$$\boxed{\bar{a}_{cm} = \frac{m_1\bar{a}_1 + m_2\bar{a}_2 + \cdots + m_n\bar{a}_n}{m_1+m_2+\cdots+m_n}}$$

$$M\bar{a}_{cm} = m_1\bar{a}_1 + m_2\bar{a}_2 + \cdots + m_n\bar{a}_n$$

from newton's second law of motion,

$$\bar{F} = m\bar{a}$$

$$M\bar{a}_{cm} = \bar{F}_1 + \bar{F}_2 + \cdots \bar{F}_n = \Sigma\bar{F}_n$$

$$\boxed{M\bar{a}_{cm} = \Sigma\bar{F}_{ext} + \Sigma\bar{F}_{internal}}$$

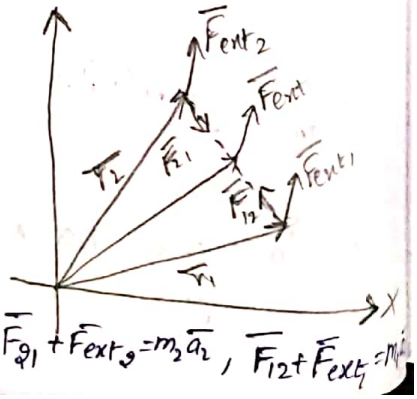
$$\bar{F}_{12} + \bar{F}_{ext_1} + \bar{F}_{21} + \bar{F}_{ext_2} = m_1\bar{a}_1 + m_2\bar{a}_2$$

from newton's third law,

action = -reaction.

i.e, $\bar{F}_{21} = -\bar{F}_{12}$

$$\bar{F}_{21} + \bar{F}_{12} = 0$$



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$\Rightarrow \quad \Sigma F_{int} = 0$

$\Sigma F_{ext} = m_1 \bar{a}_1 + m_2 \bar{a}_2$

$$\boxed{\Sigma F_{ext} = M \bar{a}_{cm}}$$

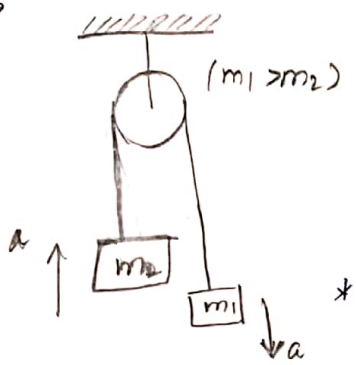
points:

1. Internal force donot effect the motion of COM.
2. Net external force appears to act at the CM of system.
3. If $\Sigma F_{ext} = 0$ then $\bar{a}_{cm} = 0$

   $\Rightarrow \bar{v}_{cm} = \text{const}$ i.e, body is either at rest (which is at rest initially) & moves continuously with const velocity.

and $\boxed{\bar{P}_{cm} = (M)_{system} \times \vec{V}_{cm} = \text{const}}$ → law of conservation of Linear momentum.

→



acceleration of blocks

$$a = \frac{(m_1 - m_2)g}{m_1 + m_2}$$

* acceleration of COM,

$$a_{COM} = \frac{m_1 \bar{a}_1 + m_2 \bar{a}_2}{m_1 + m_2}$$

$$= \frac{m_1 a - m_2 a}{m_1 + m_2}$$

$$\boxed{a_{COM} = \frac{(m_1 - m_2)^2}{(m_1 + m_2)^2} g}$$

* displacement of COM in a time t:

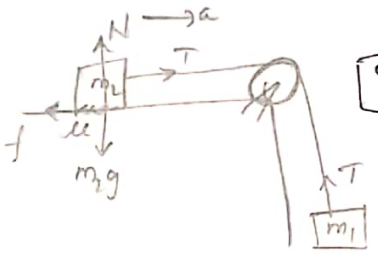
[for $u_{COM} = 0$] $\quad \boxed{S_{cm} = \frac{1}{2} a_{cm} t^2}$

* velocity of COM at time t:

$$\boxed{V_{cm} = a_{cm} t}$$

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→



$T = m_1 g$ $\quad m_1 g > \mu m_2 g$

$\boxed{m_1 > \mu m_2}$

$m_1 g - T = m_1 a$ $\quad a$ then blocks move together.

$T - \mu m_2 g = m_2 a$

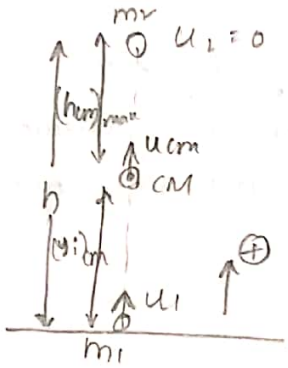
add

$$\boxed{a = \frac{(m_1 - \mu m_2) g}{m_1 + m_2}}$$

$$a_{COM} = \frac{m_1 \bar{a_1} + m_2 \bar{a_2}}{m_1 + m_2}$$

$$a_{COM} = \frac{\sqrt{m_1^2 a^2 + m_2^2 a_2^2}}{m_1 + m_2}$$

→



Find max height attained by COM of the system.

$$u_{cm} = \frac{m_1 u_1 + m_2 u_2}{m_1 + m_2}$$

$$= \frac{m_1 u_1 + m_2 (0)}{m_1 + m_2}$$

$$\boxed{u_{cm} = \frac{m_1 u_1}{m_1 + m_2} (\uparrow)}$$

Initial position of COM,

$$(y_{cm})_{(i)} = \frac{m_1 y_1 + m_2 y_2}{m_1 + m_2}$$

$$\boxed{y_{cm_{(i)}} = \frac{m_2 h}{m_1 + m_2}}$$

acceleration of COM,

$$a_{cm} = \frac{m \bar{a_1} + m_2 \bar{a_2}}{m_1 + m_2}$$

$$= \frac{m_1 (-g) + m_2 (-g)}{m_1 + m_2}$$

$$= -g$$

$$\boxed{a_{cm} = g (\downarrow)}$$

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Max height attained by COM:

$$\vec{v}_{cm} - \vec{u}_{cm} = 2 a_{cm} (h_{cm})_{max}$$

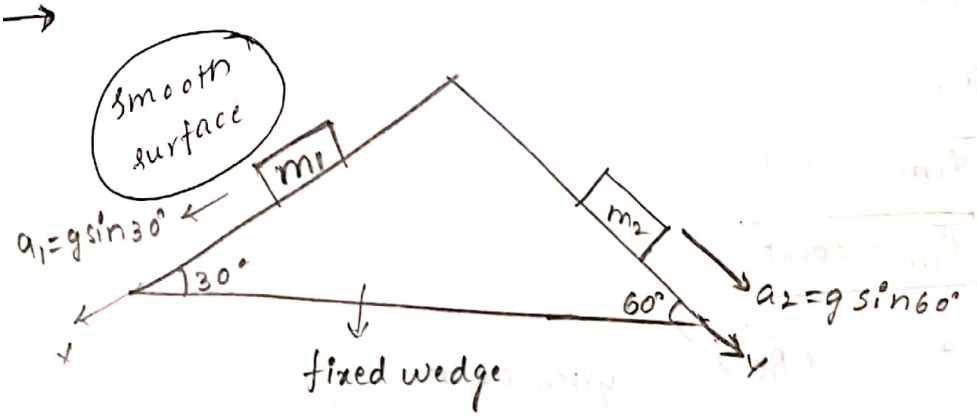
$$(h_{cm})_{max} = \frac{-\vec{u}_{cm}^{\,2}}{2a_{cm}}$$

$$= \frac{\vec{u}_{cm}^{\,2}}{2(+g)}$$

$$\boxed{(h_{cm})_{max} = \frac{\vec{u}_{cm}^{\,2}}{2g}}$$

Internal forces donot

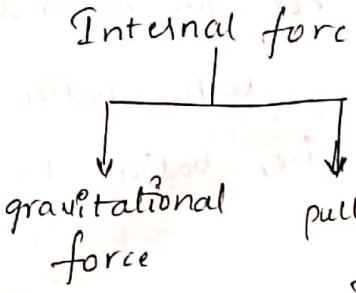
→



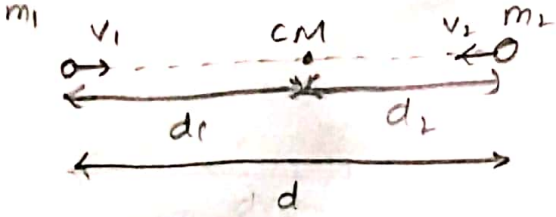
if they are connected with string, then they have common acceleration.



Internal forces donot effect the motion of CofM. & path of COM:



→



$$d_1 = \frac{m_2 d}{m_1 + m_2}, \quad d_2 = \frac{m_1 d}{m_1 + m_2}$$

* If both are moving under the influence of gravitational force (internal force)

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Here

① $v_{cm} = 0$ i.e., $\boxed{m_1 v_1 = m_2 v_2}$

* Finally they meet at their centre of mass.

* If $F_{ext} = 0 \Rightarrow a_{cm} = 0$

& $v_{cm} = const$

* $S_{cm} = v_{cm}\,(t)$

Law of Conservation of Linear Momentum:

* If $\Sigma f_{net} = 0$

$\bar{a}_{cm} = 0$

$\bar{v}_{cm} = const$

$\boxed{\bar{P}_{cm} = const}$

$\bar{P}_1 + \bar{P}_2 + \bar{P}_3 + \cdots + \bar{P}_n = (\bar{P}_f)_{system}$ const

i.e, $\boxed{(\bar{P}_i)_{system} = (\bar{P}_f)_{system}}$ *

* L.C.L.M:

① $\Sigma F_{ext} = 0$

② For two bodies system,

both bodies must be in motion;

i.e, bodies should be movable.

Applications:

① Explosion of a bomb:



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$\bar{P}_f = \bar{P}_1 + \bar{P}_2 = m_1\bar{v}_1 + m_2\bar{v}_2$

$\bar{P}_i = \bar{P}_f$

$0 = m_1v_1 + m_2v_2 \Rightarrow \boxed{m_1v_1 = -m_2v_2}$

* Energy of explosion:
(chemical energy of the bomb)

$$E = K \cdot E_f - K \cdot E_i$$

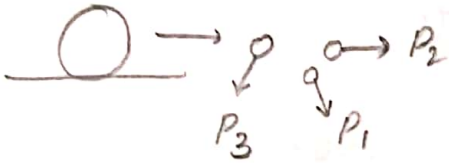
$$= \frac{1}{2}m_2v_2^2 - \frac{1}{2}m_1v_1^2 = 0$$

$$E = \frac{P^2}{2m_1} + \frac{P^2}{2m_2}$$

$$E = \frac{P^2}{2}\left(\frac{1}{m_1} + \frac{1}{m_2}\right)$$

$$E = \frac{P^2}{2}\left(\frac{m_1+m_2}{m_1m_2}\right)$$

* If bomb split into 3 pieces.



$\bar{P}_f = \bar{P}_1 + \bar{P}_2 + \bar{P}_3$

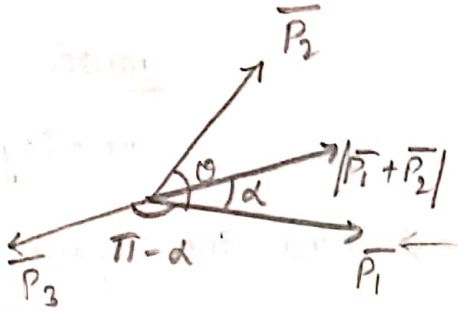
According to L.C.E.M,

$\bar{P}_f = \bar{P}_i$

$P_1 + P_2 + P_3 = 0$

$\bar{P}_3 = -(\bar{P}_1 + \bar{P}_2)$

$|\bar{P}_3| = |\bar{P}_1 + \bar{P}_2|$



$|\bar{P}_3| = \sqrt{P_2^2 + P_1^2 + 2P_1P_2\cos\theta}$

$\tan\alpha = \frac{P_2\sin\theta}{P_1 + P_2\cos\theta}$

* Energy of explosion:

$$E = \frac{P_1^2}{2m_1} + \frac{P_2^2}{2m_2} + \frac{P_3^2}{2m_3}$$

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*

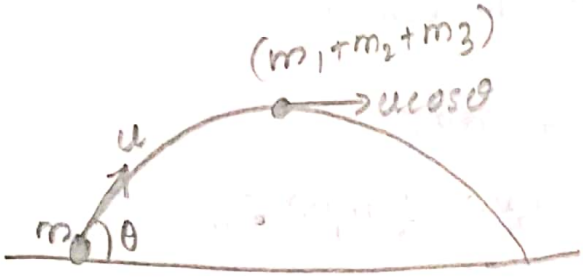


$(m_1+m_2)v\hat{j} = m_1\bar{v_1} + m_2\bar{v_2}$

bomb blasts at mid air



*



bomb explodes at the highest point.

$\bar{P_i} = \bar{P_f}$

$(m_1+m_2+m_3)\,u\cos\theta\hat{i} = m_1 u\cos\theta(-\hat{i}) + m_2(0) + m_3(\bar{v_3})$

$$x_{cm} = \frac{m_1x_1 + m_2x_2 + m_3x_3}{m_1+m_2+m_3}$$

$$R = \frac{m_1(0) + m_2\left(\frac{R}{2}\right) + m_3x_3}{m_1+m_2+m_3}$$

$$\boxed{R = \frac{u^2\sin 2\theta}{g}}$$

Firing bullet from gun:

→



Mass of gun + bullets = 50m

m = mass of each bullet

$$\frac{\text{muzzle speed of bullet}}{\text{(velocity of bullet w.r.to gun)}} = 200\,m/s$$

Find the speed of gun just after 2nd shot.

$\bar{P_f} = \bar{P_i}$

* Let V_1 be the velocity of recoil velocity of gun just after 1st shot.



$\bar{V_b} = \bar{V}_{b/g} = \bar{V_b} - \bar{V_g} \Rightarrow \bar{V_b} = \bar{V_0} + \bar{V_g}$

$= 200\hat{i} + V_1\hat{i}$

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$\overline{P_f} = \overline{P_i}$

$49mV_1 (-\uparrow) + m(200 - V_1)(\uparrow) = 0$

$49mV_1 = m(200 - V_1)$

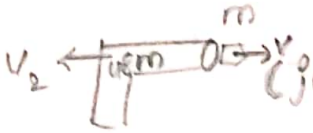
$49V_1 = 200 - V_1$

$V_1 = \frac{200}{50}$

$\boxed{V_1 = 4 \text{ m/s}}$



(just before 2nd shot)



(just after 2nd shot)

$\overline{P_i} = \overline{P_f}$

$49m.V_1(-\uparrow) = 48mV_2(-\uparrow) + m(200 - V_2)\uparrow$

$49mV_1 = 48mV_2 + m(V_2 - 200)$

$49V_1 = 48V_2 + V_2 - 200$

$49(4) = 49V_2 - 200$

$3^{49}_{\times 4}$
196

$\frac{49 \times 4 + 200}{49} = V_2$

$\frac{396}{49} = V_2$

$\boxed{V_2 = \frac{396}{49} \text{ m/s}}$

→ Figure shows woman + rocket lancher of total mass 90kg.



$u = 5 \text{ m/s}$, mass of rocket = 10kg,
muzzle velocity of rocket $(V_0) = 50 \text{ m/s}$.

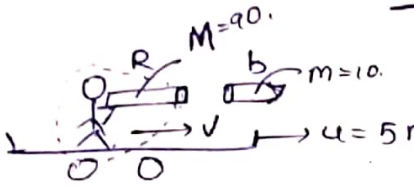
1) what will be woman's and rockets velocity after firing?

2) Find energy of explosion.

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$\bar{V}_b = \bar{V}_{b/g} \pm V_g - V_g \Rightarrow \bar{V}_b = V_0 + V_g$ *

$\bar{V}_b = (V_0 + V_g)\uparrow$



$\bar{P}_i = \bar{P}_f$

$V_{b/R} = 30.$

$V_b = V_{b/R} + V_R$

$= 30 + 5 = 35$ m/s

$(90+10)5 = 10 \times 35 + 90(V)$

$500 = 350 + 90(V)$

$150 = 90(V)$

$V = \frac{5}{3}$ m/s.

E. of Explosion = $K.E_f - KE_i$

$= \frac{1}{2} \times m \times V_b^2 + \frac{1}{2} M V^2 - \frac{1}{2}(M+m)v^2$

$= \frac{1}{2} \times 10 \times (35)^2 + \frac{1}{2} 90 \times \left(\frac{5}{3}\right)^2 - \frac{1}{2}(90+10)^{25}$

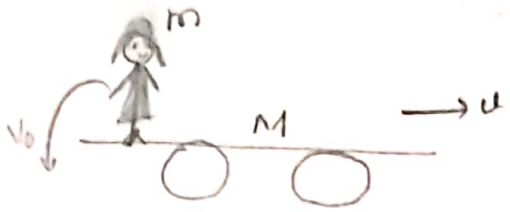
$= 6125 + 125 - 1250.$

$= 6250 - 1250$

$= 5000$ J

ans: 4050 J

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$$(m+M)u\uparrow = Mv_1\uparrow + m(v_0 - v_1)(-\uparrow)$$

$$\bar{v}_m = \bar{v}_{m/M} + \bar{v}_M$$

$$\bar{v}_m = v_0(-\uparrow) + v_1(\uparrow)$$
$$= (v_1 - v_0)(\uparrow)$$



→ mass of man = 80kg (m_1), Trolley mass (m_2) = 40 kg. Man jumps off the back of the trolley so that his relative ~~velo~~ speed to the ground is 2 m/s in the direction opposite to the direction of motion of trolley.

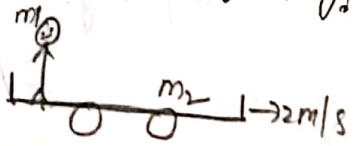
Find ① speed of C.M of man-trolley system before & after his jumps?

② what is the speed of trolley just after the man jumps?

③ what is the speed of C.M of the system after man hits the ground & comes to rest?

④ How much energy does the man expend in jumping?

① $v_{cm} = \dfrac{m_1v_1 + m_2v_2}{m_1+m_2}$, $v_1 = v_2 = 2$ m/s



$$= \frac{(m_1+m_2)2}{m_1+m_2} = 2 \text{ m/s (just before jumping)}$$

friction b/w feet & trolley is responsible for jumping and is the internal force of system which doesn't change the speed of C.M.

Velocity of C.M after jumping = 2 m/s.

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(2) $\Sigma F_{ext} = 0$ (along horizontal)

$$\bar{P}_i = \bar{P}_f$$

$$(m_1+m_2)\,2 = m_1 v_1(-1) + m_2 v_2$$

$$(80+40)\,2 = -80 + 40(v_2)$$

$$v_2 = \frac{320}{40} = 8\ m/s.$$

(3) $$v_{cm} = \frac{m_1 v_1 + m_2 v_2}{m_1+m_2}$$

$$= \frac{80(0) + 40(8)}{80+40}$$

$$= \frac{320}{120} = 2.66\ m/s$$

* C.M of system change because of friction acts b/w feet & ground as external force.

(4) friction b/w feet & trolley doesn't change motion of CM but internal energy of man is utilized for jumping.

$$E = K.E_f - K.E_i$$

$$= \frac{1}{2}m_1 v_1^2 + \frac{1}{2}m_2 v_2^2 - \frac{1}{2}(m_1+m_2)v^2$$

$$= \frac{1}{2}(80)(1)^2 + \frac{1}{2}(40)(4)^2 - \frac{1}{2}(120)(2)^2$$

$$= \frac{1}{2}\times 80 + \frac{1}{2}\times 40\times 64 - \frac{1}{2}\times 120\times 4$$

$$= 40 + 1280 - 240$$

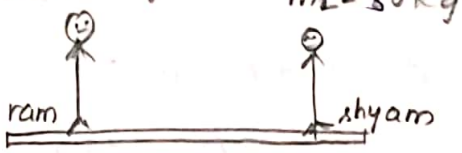
$$= 1280 - 200$$

$$\boxed{E = 1080\ J}$$

32 × 4 / 128

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Both of them can jump with a velocity of 1 m/s relative to the plank. Find the velocity of plank after the event



plank mass (m_3) = 20kg

① Rama alone jump to the left — 3/8 m/s

② sham alone jump to the right — -1/2 m/s

③ Ram jumps to left and sham jumps to right simultaneously — -1/8 m/s

④ Ram jumps to left and after that sham jumps to right — $-\frac{17}{40}$ m/s

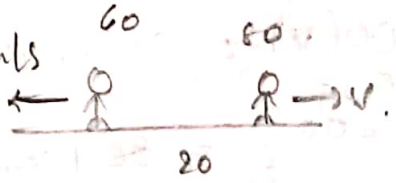
①

$$0 = (m_2+m_3)v_1(\uparrow) + m_1(\bar{v}_2)$$

$$v_2 = v_m = \bar{v}_{m/p} + \bar{v}_p$$

$$= 1(-\hat{i}) + v_1(\uparrow)$$

$$= (v_1 - 1)\hat{i}$$



$$0 = (80+20)v + 60(v-1)$$

$$100v = 60 - 60v$$

$$160v = 60$$

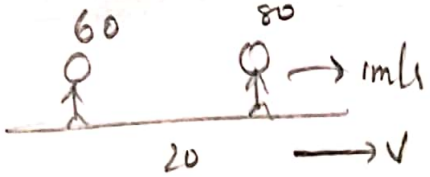
$$\boxed{v = 3/8 \text{ m/s}}$$

②

$$0 = 80(v) + 80(1+v)$$

$$0 = 2v + 1$$

$$\boxed{v = -1/2 \text{ m/s}}$$



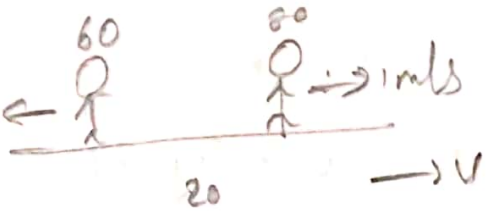
$$v_{S/T} = 1 \text{ m/s}$$

$$\bar{v}_S = \bar{v}_{S/T} + \bar{v}_T$$

$$= 1 + v$$

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(3)



$$0 = 1\phi(v) + 3\phi(v-1) + 4\phi(v+1)$$

$$0 = v + 3v - 3 + 4v + 4$$

$$0 = 8v + 1$$

$$\boxed{v = -1/8 \text{ m/s}}$$

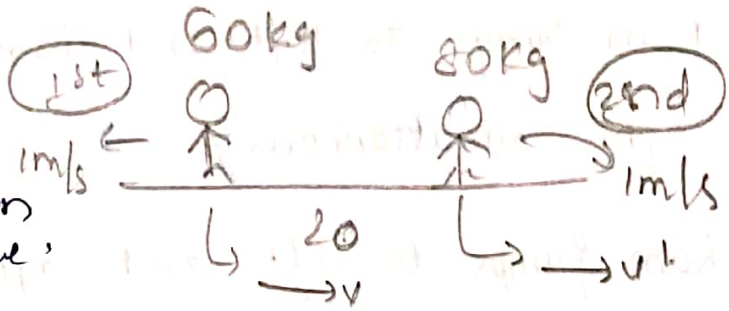
$V_{R/T} = -1$

$V_R = -1 + V$

$V_{S/T} = 1$

$V_S = 1 + V$

(4)



$$0 = 60(v-1) + 100v$$

$v = 3/8$ → ① question one,

$$P_i = P_f \quad \text{(after ram jumped)}$$

$$100(v) = 80(1+v1) + 20(v1)$$

$$100 \times \frac{3}{8} = 80 + 100v1$$

$$\frac{15}{2} - 16 = 20v1$$

$$\frac{15 - 32}{2} = 20v1$$

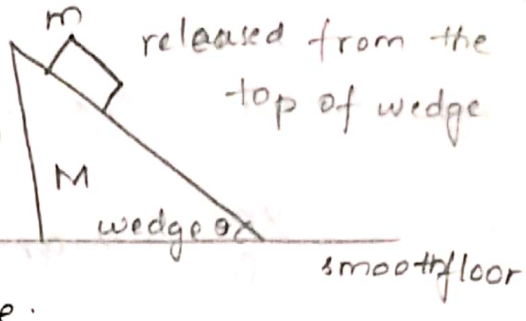
$$\frac{-17}{2 \times 20} = v1$$

$$\boxed{v1 = -17/40 \text{ m/s}}$$

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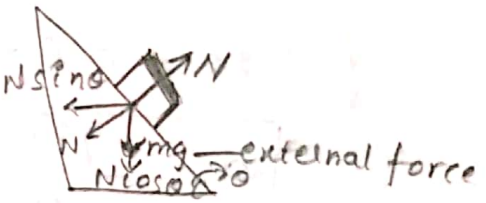
→ Find

① displacement of wedge on the horizontal ground when the block reaches the bottom of the wedge.



② Find acceleration of CM of the system.

① since there is no external force acting along horizontal.



Along horizontal,

$$\overline{P_i} = \overline{P_f}$$

$$0 = \overline{P_1} + \overline{P_2}$$



$$0 = m[u\cos\theta - v]\uparrow + M(V)(-\uparrow)$$

$$(m+M)V = mu\cos\theta$$

$$\boxed{V = \frac{mu\cos\theta}{m+M}}$$

$$u = \frac{AB}{t} = \frac{L}{\cos\theta \times t}$$

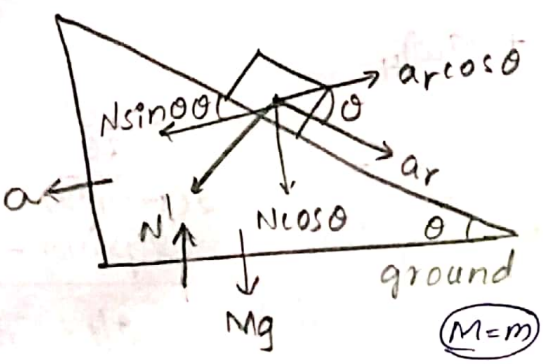
$$\cot\theta = \frac{L}{b}$$

$$\boxed{L = h\cot\theta}$$

$$\frac{x_\omega}{t} = \frac{mu\cos\theta}{M+m} \Rightarrow \frac{x_\omega}{t} = \frac{\frac{mL}{\cos\theta \times t} \times \cos\theta}{M+m}$$

$$\boxed{x_\omega = \frac{mL}{M+m}}$$ → relative displacement along horizontal direction.

②



(M=m)

$$N_1 = N\cos\theta + mg$$

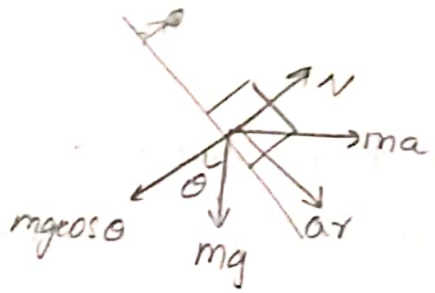
$$N\sin\theta = ma \quad \text{—①}$$

$$Ma = m(a\cos\theta - a)$$

$$a = a\cos\theta \quad \text{—②}$$

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on block,



$$mg\cos\theta = N + ma\sin\theta \quad -③$$

$$mg\cos\theta = \frac{ma}{\sin\theta} + ma\sin\theta$$

$$g\cos\theta = \frac{a(1+\sin^2\theta)}{\sin\theta}$$

$$\frac{g}{2}\sin 2\theta = a(1+\sin^2\theta)$$

$$\boxed{a = \frac{g\sin 2\theta}{2(1+\sin^2\theta)}} \rightarrow \text{acceleration of wedge.}$$

2 ⇒

$$a_r = \frac{2a}{\cos\theta} = \frac{2g\sin\theta\cos\theta}{(1+\sin^2\theta)(\cos\theta)} = \frac{2g\sin\theta}{1+\sin^2\theta}$$

relative acceleration of block w.r.to wedge.

Absolute acceleration of block along vertical:

$$(\bar{a}_b)_v = (\bar{a}_{b/w})_v + (\bar{a}_w)_v$$

$$(a_b)_v = (a_{rel})_v$$

$$= a_r\sin\theta = \frac{2g\sin\theta}{1+\sin^2\theta}\cdot\sin\theta$$

$$(\bar{a}_b)_v = \frac{2g\sin^2\theta}{1+\sin^2\theta}$$

Absolute acceleration of block along horizontal:

$$(a_b)_H = (a_{rel})_H + (a_w)_H$$

$$= a_r\cos\theta + (-a) = a = \frac{g\sin 2\theta}{2(1+\sin^2\theta)}$$

$$= 2a - a = a$$

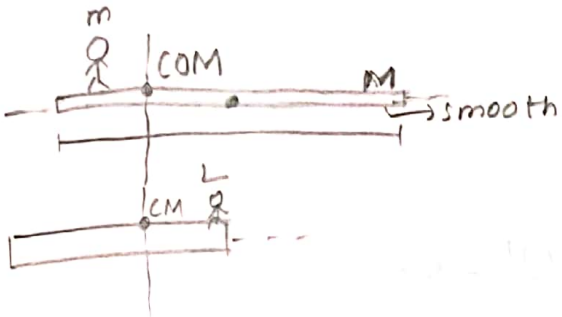
$$= \frac{g\sin\theta}{1+\sin^2\theta} - \frac{g\sin 2\theta}{2(1+\sin^2\theta)} = \frac{2g\sin\theta - g\sin 2\theta}{2(1+\sin^2\theta)}$$

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$$a_{cm} = \frac{m(a_{co}) + m(a_b)_v}{m+m}$$

$$= \frac{(a_b)_v}{2} = \frac{g\sin^2\theta}{1+\sin^2\theta}$$

→ Boat-man Problems-



M-I :-

* displacement of man on plank,

$$\boxed{d_{rel} = L\uparrow}$$

* displacement of plank = x (to left)

Here C.M of system is fixed.

$$\Delta x_{cm} = 0$$

$$m_1\vec{s}_1 + m_2\vec{s}_2 = 0$$

$$Mx(-\uparrow) + m(L-x)\uparrow = 0$$

$$Mx = m(L-x)$$

$$Mx = mL - mx$$

$$\boxed{x = \frac{mL}{M+m}} \rightarrow \text{displacement of planek.}$$

* absolute displacement of man = $L - x = L - \frac{mL}{M+m}$

$$\boxed{x_{am} = \frac{ML}{M+m}}$$

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Method-2 :-

$$(x_{cm})_i = (x_{cm})_f \quad -①$$

from fig,

$$(x_{cm})_i = \frac{m(0) + M\left(\frac{L}{2}\right)}{m+M} = \frac{ML}{2(m+M)}$$

& $$(x_{cm})_f = \frac{m(L-x) + M\left(\frac{L}{2} - x\right)}{M+m}$$

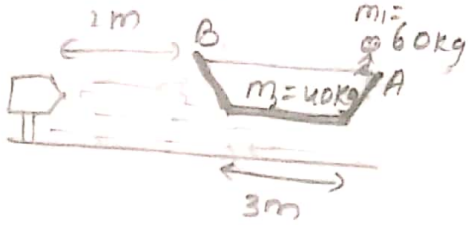
$$\frac{ML}{2(m+M)} = \frac{mL - mx + \frac{ML}{2} - Mx}{M+m}$$

$$\frac{ML}{2} = (2m+M)\frac{L}{2} - (m+M)x$$

$$\frac{Mx}{2} - mL - \frac{Mx}{2} = -(m+M)x$$

$$mL = (M+m)x$$

$$\boxed{x = \frac{mL}{M+m}}$$



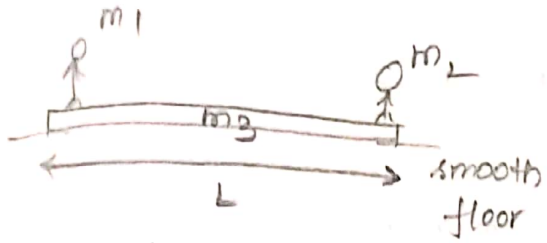
when man walks from A to B Find distance of man from shore?

* displacement of bloat,

$$x = \frac{36 \times 3}{100} = \frac{9}{5} = 1.8\text{m} \text{ (to right)}$$

* distance of man from shore $= 2 + 1.8 = 3.8\text{m}$.

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when both the mass interchange.

Here $S_{cm} = 0$

$$\frac{m_1\bar{s}_1 + m_2\bar{s}_2 + m_3\bar{s}_3}{m_1+m_2+m_3} = 0$$

$$m_1(L-x)\uparrow + m_2(L+x)(-\uparrow) + m_3 x(-\uparrow) = 0$$

$$m_1(L-x) = m_2(L+x) + m_3 x$$

$$m_1 L - m_1 x = m_2 L + m_2 x + m_3 x$$

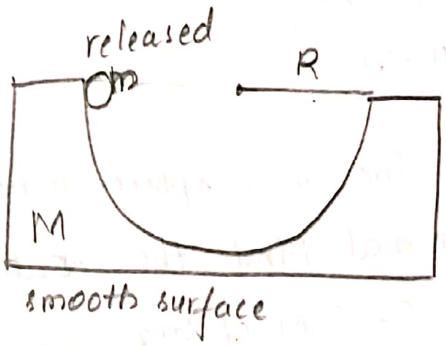
$$\boxed{\frac{(m_1-m_2)L}{m_1+m_2+m_3} = x}$$

$\bar{S}_m = \bar{S}_{m/p} + \bar{S}_p$

$= -L - x$

$= -(L+x)$

$= (L+x)(-\uparrow)$



since there is no F_{ext} along horizontal.

$\bar{P}_f = \bar{P}_i$

V_1, V_2 — absolute velocities.



here wedge has max speed when ball is at the bottom most point

$0 = mV_1 - MV_2$

$mV_1 = MV_2 \quad -(1)$

& $\bar{V}_{rel} = \bar{V}_1 - \bar{V}_2$

$$\boxed{\bar{V}_{rel} = (V_1+V_2)\uparrow}$$

we use L.C.E,

$$mgR = mgr + \frac{1}{2}mV_1^2 + \frac{1}{2}MV_2^2$$

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$$mg(R-r) = \frac{1}{2}m(v_1)^2 + \frac{1}{2}Mv_2^2 \quad -(2)$$

using eq (1) & (2), we find v_1, v_2.

$$mg(R-r) = \frac{1}{2}m(v_1)^2 + \frac{1}{2}\frac{m^2v_1^2}{M}$$

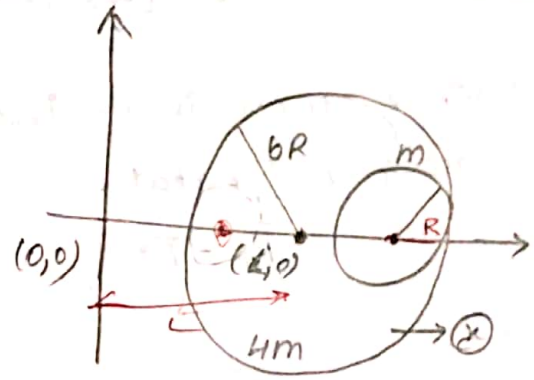
$$mg\,g(R-r) = \frac{v_1^2}{2}\left(m_1 + \frac{m}{M}\right)$$

$$\frac{2g(R-r)(M)}{(M+m)} = v_1^2$$

$$\boxed{v_1 = \sqrt{\frac{2g(R-r)(M)}{(M+m)}}}$$

Let x be the displacement of wedge,

$$\boxed{x = \frac{m(R-r)}{m+M}}$$



The small sphere is now released. Find the co-ordinates of COM of large sphere when small sphere reaches the other extreme position

Method - 1 :-

$$x = \frac{M(12R-2R)}{4m+m} = \frac{10Rm}{5m} = 2R$$

Method - 2 :-

$$(x_{cm})_i = (x_{cm})_f$$

$$\frac{M(L+5R) + 4m(L)}{5m} = \frac{4m(L+x) + m(L+x-5R)}{5m}$$

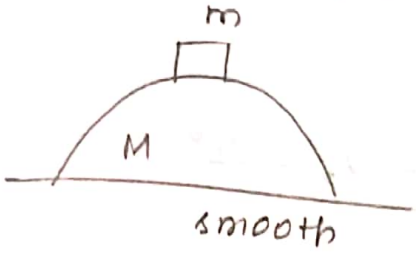
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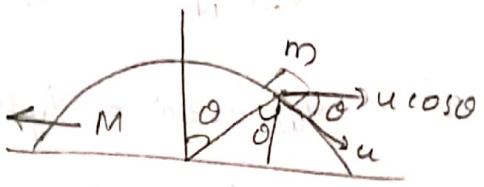
$$x + 5R + 4x = 4x + 4x + x + x - 5R$$

$$5R = 5x - 5R$$

$$R + R = x \Rightarrow \boxed{x = 2R}$$

→





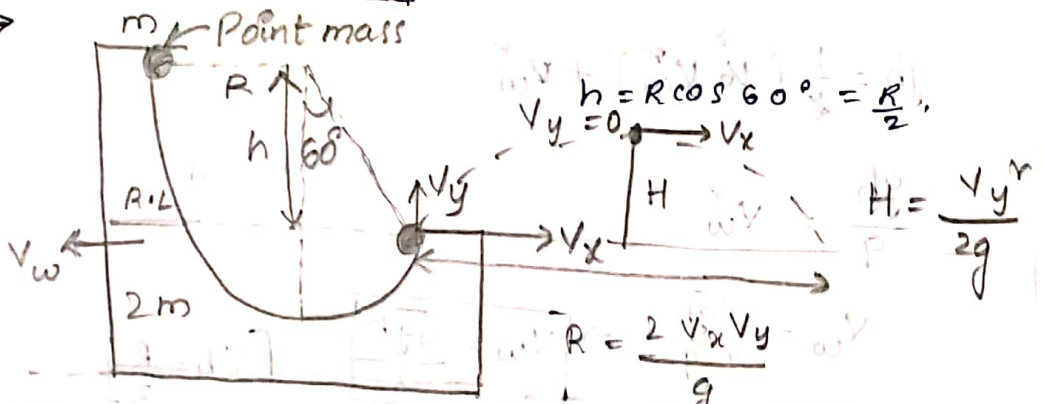
$$MV = m(u\cos\theta - V)$$

$$MV = -mV + mu\cos\theta$$

$$\boxed{V = \frac{mu\cos\theta}{m+M}}$$

Q $\boxed{x = \frac{mR\sin\theta}{m+M}}$

→



all surfaces are smooth

① L.C.L.M, (along $F_{net} = 0$)

$\vec{P}_i = \vec{P}_f$ (along horizontal)

$0 = m \cdot v_x \uparrow + 2m v_w (-\uparrow)$

$v_x = 2v_w$ — ①

② h.C.E,

$mgh = \frac{1}{2}m(v_x^2 + v_y^2) + \frac{1}{2}(2m)v_w^2$

$gh = \frac{1}{2}(v_x^2 + v_y^2) + v_w^2$ — ②

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③ Constraint eqn:

Along normal to contact surface,

$$\boxed{\text{relative velocity} = 0}$$

$$(V_\omega)_N = (V_b)_N$$

$$V_\omega \cos 30^\circ = V_y \cos 60^\circ - V_x \cos 30^\circ$$

$$\frac{\sqrt{3}V_\omega}{2} = \frac{V_y}{2} - \frac{\sqrt{3}V_x}{2}$$

$$\sqrt{3}(V_\omega + V_x) = V_y$$

$$\boxed{2\sqrt{3}V_\omega = V_y} \quad -③$$

①, ③ in ②

$$gh = \frac{1}{2}(4V_\omega^2 + 4\times 3V_\omega^2) + V_\omega^2$$

$$gh = \frac{1}{2}(16V_\omega^2) + V_\omega^2$$

$$\frac{gh}{9} = V_\omega^2$$

$$V_\omega = \sqrt{\frac{gh}{9}} \Rightarrow \boxed{V_\omega = \frac{\sqrt{gh}}{3}} \quad [\because h = \frac{R}{2}]$$

or $(h = R\cos\theta)$

$$\boxed{V_x = \frac{2\sqrt{gh}}{3}}, \quad \boxed{V_y = 2\sqrt{\frac{gh}{3}}}$$

* Speed of ball

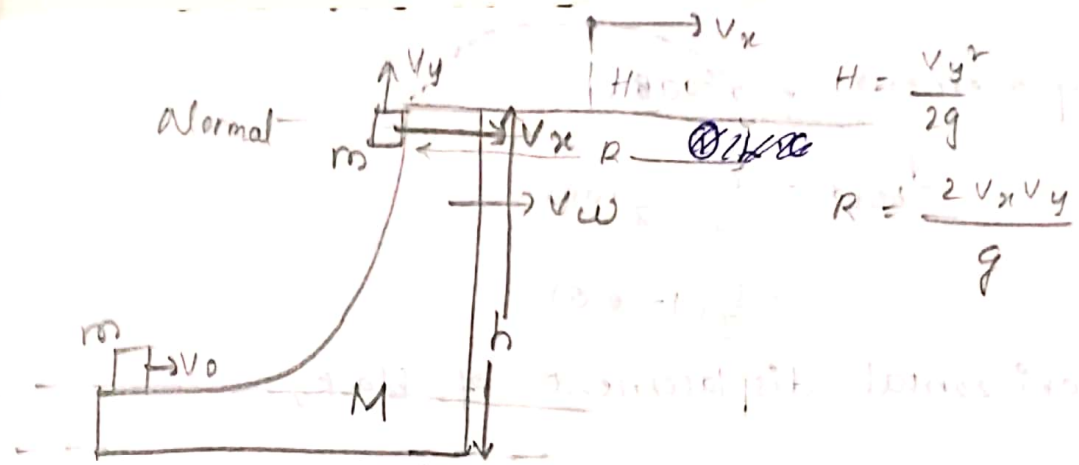
$$V_b = \sqrt{V_x^2 + V_y^2}$$

$$= \sqrt{\frac{4gh}{9} + \frac{4gh}{3}}$$

$$= \frac{2}{3}\sqrt{gh + 3gh} = \frac{4}{3}\sqrt{gh}$$

$$\boxed{V_b = \frac{4}{3}\sqrt{gh}}$$

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All surfaces are smooth

① L·C·L·M, (along horizontal)

$\vec{P}_i = \vec{P}_f$

$\Rightarrow mV_0\uparrow = mV_x\uparrow + MV_\omega\uparrow$

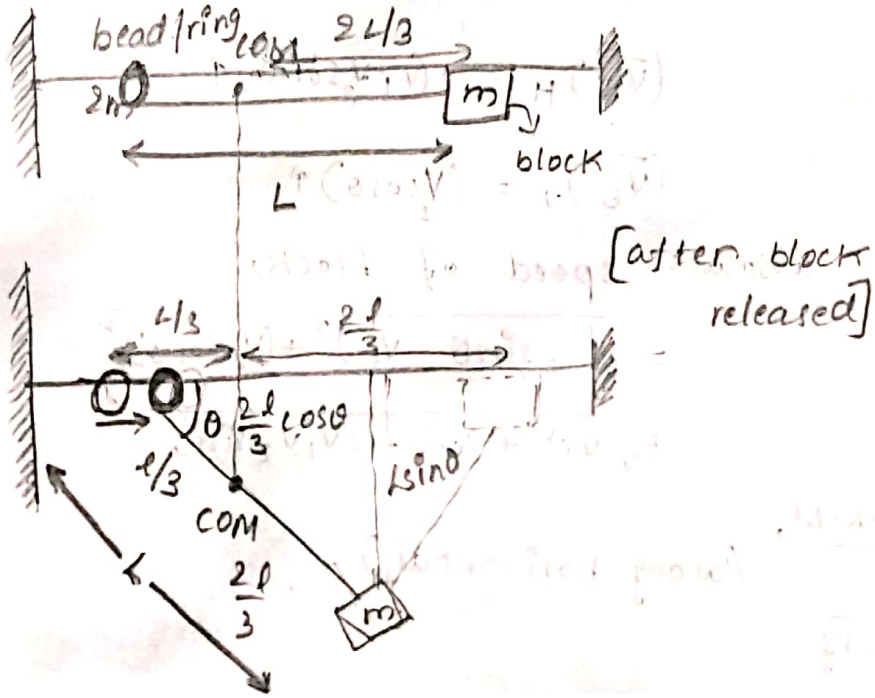
$* \quad mV_0 = mV_x + MV_\omega$ — ①

② L·C·E,

$\frac{1}{2}mV_0^2 = mgh + \frac{1}{2}m(V_x^2 + V_y^2) + \frac{1}{2}MV_\omega^2$ — ③

③ constraint eqn:

$V_\omega = V_x$



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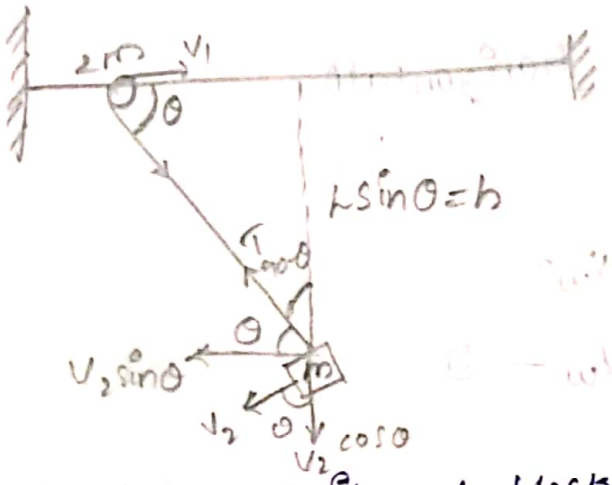
* displacement of bead,

$$d_{bead} = \frac{L}{3} - \frac{L}{3}\cos\theta$$

$$= \frac{L}{3}(1-\cos\theta)$$

* horizontal displacement of block,

$$d_{block} = \frac{2L}{3} - \frac{2L}{3}\cos\theta = \frac{2L}{3}(1-\cos\theta)$$



V_1 = speed of bead

V_2 = speed of block w.r.to bead.

* Absolute velocity of block,

$$V_{block} = V_{block/bead} + V_{bead}$$

$$\overline{V_b} = \overline{V_2} + \overline{V_1}$$

$$\& \quad (\overline{V_b})_H = (-V_2\sin\theta + V_1)\uparrow$$

$$(\overline{V_b})_H = (V_1 - V_2\sin\theta)\uparrow$$

$$(\overline{V_b})_V = (V_2\cos\theta)\uparrow$$

* Net absolute speed of block,

$$= \sqrt{(V_2\sin\theta - V_1)^2 + (V_2\cos\theta)^2}$$

$$= \sqrt{V_1^2 + V_2^2 - 2V_1V_2\sin\theta}$$

* L.C.L.M, (along horizontal)

$$\overline{P_i} = \overline{P_f}$$

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$$0 = 2mv_1 \uparrow + m(v_2 \sin\theta - v_1)(-\uparrow)$$

$$2mv_1 = m(v_2 \sin\theta - v_1)$$

$$3v_1 = v_2 \sin\theta \quad \text{—①}$$

* h.C.E,

$$0 = -mgh + \frac{1}{2}(2m)(v_1)^2 + \frac{1}{2}(m)(v_2^2 + v_1^2 - 2v_1v_2\sin\theta$$

$$mgh = \frac{1}{2}(2m)(v_1^2) + \frac{1}{2}m(v_1^2 + v_2^2 - 2v_1v_2\sin\theta)$$

$$2gh = 2v_1^2 + v_1^2 + v_2^2 - 2v_1(3v_1)$$

$$2gh = 3v_1^2 - 6v_1^2 + v_2^2$$

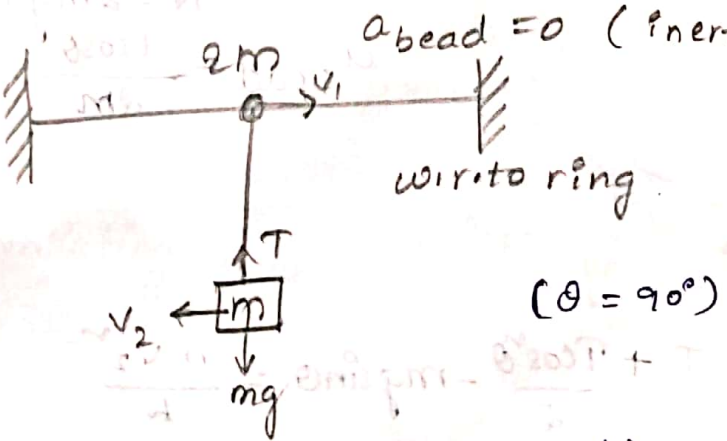
$$2gh = v_2^2 - 3v_1^2$$

$$2gL\sin\theta = v_2^2 - \frac{v_2^2\sin^2\theta}{3}$$

$$6gL\sin\theta = 3v_2^2 - v_2^2\sin^2\theta$$

$$\boxed{v_2^2 = \frac{6gL\sin\theta}{3 - \sin^2\theta}}$$

** Tension is the string when string is vertical:



$$0 = 2mv_1 \uparrow + m(v_2 - v_1)(-\uparrow)$$

$$2mv_1 = m(v_2 - v_1)$$

$$3v_1 = v_2 \quad \text{—①}$$

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$$T - mg = \frac{mv_2^2}{L} \quad -(2)$$

* L.C.E,

$$0 = -mgL + \frac{1}{2}(2m)(v_1)^2 + \frac{1}{2}m(v_2 - v_1)^2$$

$$2gL = 2\left(\frac{v_2}{3}\right)^2 + \left(\frac{2v_2}{3}\right)^2$$

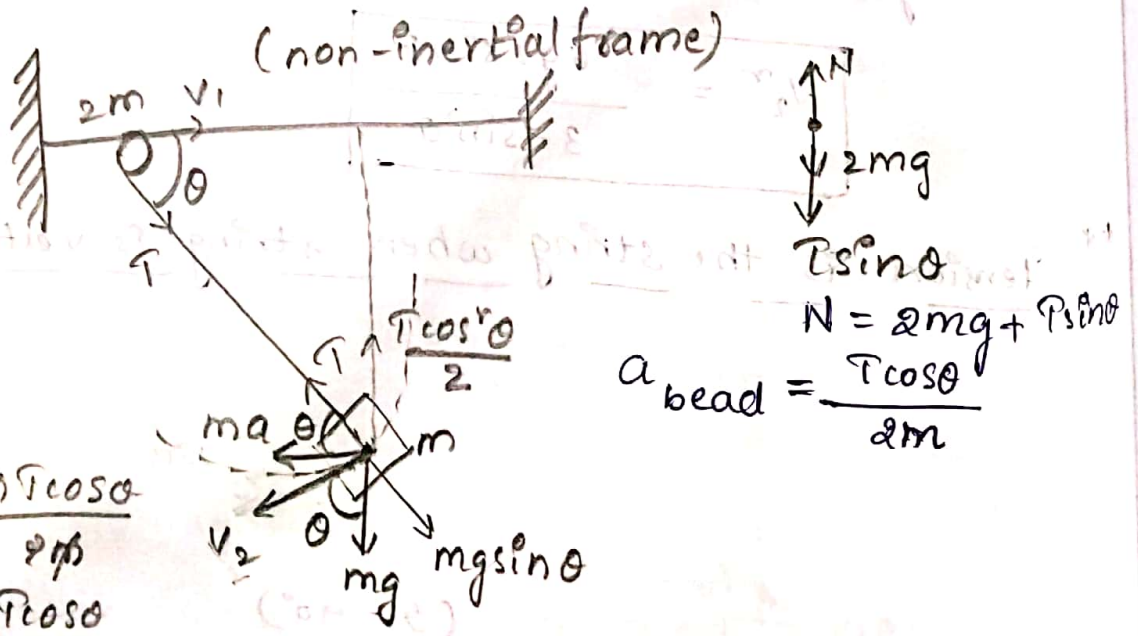
$$2gL = \frac{2v_2^2}{9} + \frac{4v_2^2}{9}$$

$$3(2)gL = 6v_2^2$$

$$v_2^2 = 3gL$$

$$T - mg = \frac{m(3gL)}{L}$$

$$\boxed{T = 4mg}$$

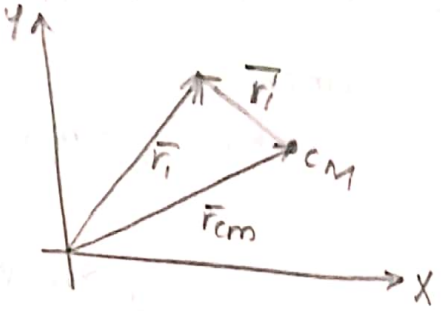


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Centre of mass frame (c-frame):

If origin of co-ordinate-system is fixed at C.M of system then it is called C.M frame.

From figure,



$$\boxed{\bar{r}_{cm} + \bar{r}_i' = \bar{r}_i} \quad \text{— (a)}$$

we have,

$$\bar{r}_{cm} = \frac{1}{M} \Sigma m_i \bar{r}_i$$

If origin is fixed at centre of mass of system then $\boxed{\bar{r}_{cm} = 0}$

$$\Rightarrow \quad \Sigma m_i r_i = 0$$

$$\Sigma m_i r_i^d = 0$$

on differentiating,

$$\Sigma m_i \bar{v}_i' = 0$$

$\bar{v}_i'$ = velocity of ith particle w.r.t to CM.

$$\boxed{\bar{P}_{system\,|\,cm} = 0}^{*}$$

* centre of mass frame is called "zero momentum frame".

differentiating eqn (1),

$$\bar{v}_i = \bar{v}_i' + \bar{v}_{cm}$$

now, K.E of ith particle

$$K.E_i = \frac{1}{2} m_i |\bar{v}_i'|^2$$

$$= \frac{1}{2} m_i (\bar{v}_i' \cdot \bar{v}_i')$$

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$$= \frac{1}{2} m_i (\bar{v}_i' + \bar{V}_{cm}) \cdot (\bar{v}_i' + \bar{V}_{cm})$$

$$= \frac{1}{2} m_i \left[v_i'^2 + V_{cm}^2 + 2 \bar{v}_i' \bar{V}_{cm} \right]$$

* Total kinetic energy of all particles (in laboratory (ground frame) frame)

$$K.E_T = \frac{1}{2} \sum m_i v_i'^2 + \frac{1}{2} \sum m_i V_{cm}^2 + \frac{1}{2} \sum m_i \bar{v}_i' * \bar{V}_{cm}$$

$$= \frac{1}{2} \sum m_i v_i'^2 + \frac{1}{2} M V_{cm}^2 + \bar{V}_{cm} \sum m_i v_i' \quad (\sum m_i v_i' \to 0)$$

$$\boxed{K.E_T = K.E_{system/cm} + K.E_{cm}}$$

↳ internal K.E.

* For two particle system,

having masses m_1, m_2 moving with velocities $\bar{v}_1$ & $\bar{v}_2$ (in ground frame)

then ① $\bar{V}_{cm} = \dfrac{m_1 \bar{v}_1 + m_2 \bar{v}_2}{m_1 + m_2}$

② velocities of particles w.r.to C.M,

$$\bar{v}_{1,cm} = \bar{v}_1 - \bar{V}_{cm}$$

$$= \bar{v}_1 - \left(\frac{m_1 \bar{v}_1 + m_2 \bar{v}_2}{m_1 + m_2} \right)$$

$$= \frac{m_1 \bar{v}_1 + m_2 \bar{v}_1 - m_1 \bar{v}_1 - m_2 \bar{v}_2}{m_1 + m_2}$$

$$\boxed{\bar{v}_{1,cm} = \frac{m_2 (\bar{v}_1 - \bar{v}_2)}{m_1 + m_2}}$$

& $\bar{v}_{2,cm} = \bar{v}_2 - \bar{V}_{cm}$

$$= \bar{v}_2 - \left(\frac{m_1 \bar{v}_1 + m_2 \bar{v}_2}{m_1 + m_2} \right) = \frac{m_1 \bar{v}_2 + m_2 \bar{v}_2 - m_1 \bar{v}_1 - m_2 \bar{v}_2}{m_1 + m_2}$$

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$$\bar{V}_{2,cm} = \frac{m_1(\bar{V}_2 - \bar{V}_1)}{m_1+m_2}$$

* momentum of particles in COM-frame:

$$\bar{P}_{1,cm} = (m_1)\bar{V}_{1,cm}$$

$$= \frac{m_1 m_2 (\bar{V}_1 - \bar{V}_2)}{m_1+m_2}$$

$\bar{V}_1 - \bar{V}_2$ = relative velocity of particles ($\bar{V}_{12}$)

$$= \mu(\bar{V}_1 - \bar{V}_2) \quad ; \quad \mu = \text{reduced mass} = \frac{m_1 m_2}{m_1+m_2}$$

for n-particle system,

$$\frac{1}{\mu} = \frac{1}{m_1} + \frac{1}{m_2} + \text{- - - - -} + \frac{1}{m_n}$$

& $\bar{P}_{2,cm} = (m_2)\bar{V}_{2,cm}$

$$= \frac{m_1 m_2 (\bar{V}_2 - \bar{V}_1)}{m_1+m_2}$$

$$= \mu(\bar{V}_2 - \bar{V}_1)$$

$$\bar{P}_{1,cm} + \bar{P}_{2,cm} = \mu\bar{V}_1 - \mu\bar{V}_2 + \mu\bar{V}_2 - \mu\bar{V}_1$$

$$\boxed{\bar{P}_{system|cm} = 0}$$

* K.E system w.r.to COM:

$$K.E_{system|cm} = \frac{1}{2} m_1 V_{1,cm}^2 + \frac{1}{2} m_2 V_{2,cm}^2$$

$$= \frac{1}{2}\frac{m_1 m_2^2}{(m_1+m_2)^2}|\bar{V}_1 - \bar{V}_2|^2 + \frac{1}{2}\frac{m_1^2 m_2}{(m_1+m_2)^2}|\bar{V}_1 - \bar{V}_2|^2$$

$$= \frac{1}{2}\frac{m_1 m_2}{(m_1+m_2)}|\bar{V}_1 - \bar{V}_2|^2 \quad [m_2 + m_1]$$

$$\boxed{K_{system} = \frac{1}{2} \times \frac{m_1 m_2}{m_1+m_2}|\bar{V}_1 - \bar{V}_2|^2}$$

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$$K.E_{sy/cm} = \frac{1}{2}\mu |V_{rel}|^2$$

$$K.E_{sys\,T} = K.E_{sy/cm} + K.E_{cm}$$

* work energy theorem in COM frame:

$$W_{all\ forces} = \Delta K_{system/cm}$$

$$W_{all\ force} = \frac{1}{2}\mu \left[V_{rel}^2 - u_{rel}^2 \right]$$

$$\downarrow$$

$$W_{conservative} + W_{non\text{-}conservation}$$

Problems:

① Find maximum elongation in spring.



we solve this problem from COM frame.

$$a_c = \frac{F}{m_1+m_2}, \text{ to right}$$



* w.r.to C.M,

net force on m_1,

$$F_1' = F - m_1 a_c$$

$$= F - \frac{m_1 F}{m_1+m_2}$$

$$= \frac{m_1 F + m_2 F - m_1 F}{m_1+m_2}$$

$$\boxed{F_1' = \frac{m_2 F}{m_1+m_2}}$$

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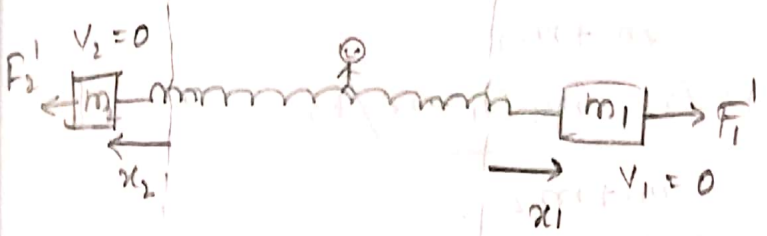
net force on m_2,

$$F_2' = m_2 a_c$$

$$F_2' = \frac{m_2 F}{m_1 + m_2}$$



w.r.to COM frame blocks are in osillatory motion



$$W_{all} = \Delta K_{system/cm}$$

$$\Rightarrow F_2' x_2 + F_1' x_1 - \frac{1}{2} k (x_1 + x_2)^2 = \frac{1}{2} \mu (v_{rel}^2 - u_{rel}^2)$$

$$\frac{m_2 F x_2}{m_1 + m_2} + \frac{m_2 F x_1}{m_1 + m_2} = \frac{1}{2} k (x_1 + x_2)^2$$

$$\frac{m_2 F}{m_1 + m_2} = \frac{1}{2} k (x_1 + x_2)$$

$$\boxed{x_{max} = x_1 + x_2 = \frac{2 m_2 F}{(m_1 + m_2) k}}$$

→ Find max elongation is spring $(F_1 > F_2)$.



$$a_c = \frac{F_1 - F_2}{m_1 + m_2}$$ **

net force on m_1,

$$F_1' = F_1 - m_1 a_c = F_1 - \frac{m_1 F_1 + m_1 F_2}{m_1 + m_2}$$

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$$= \frac{m_1F_1 + m_2F_1 - m_1F_1 + m_1F_2}{m_1+m_2}$$

$$\boxed{F_{21}' = \frac{m_1F_2 + m_2F_1}{m_1+m_2}}$$

$$F_2' = F_2 + m_2 a_L$$

$$= F_2 + \frac{m_2(F_1 - F_2)}{m_1+m_2}$$

$$= \frac{m_1F_2 + m_2F_2 + m_2F_1 - m_2F_2}{m_1+m_2}$$

$$\boxed{F_2' = \frac{m_1F_2 + m_2F_1}{m_1+m_2}}$$

$$(F_2')x_2 + (F_1')x_1 - \frac{1}{2}K(x_1+x_2)^2 = \frac{1}{2}\mu(v_{rel}^2 - u_{rel}^2)$$

$$(x_1+x_2)\left(\frac{m_1F_2 + m_2F_1}{m_1+m_2}\right) = \frac{K}{2}(x_1+x_2)^2$$

$$\frac{2}{K}\left[\frac{m_1F_2 + m_2F_1}{m_1+m_2}\right] = x_1+x_2 = x_{max}$$

$$\boxed{x_{max} = \frac{2}{K}\left[\frac{m_1F_2 + m_2F_1}{m_1+m_2}\right]}$$



① Find the position of COM at time 't'.

② If the extension of the spring is x_0 at time 't'. Find displacements of blocks at

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that instant.

① $a_c = \frac{F}{m_1+m_2}$

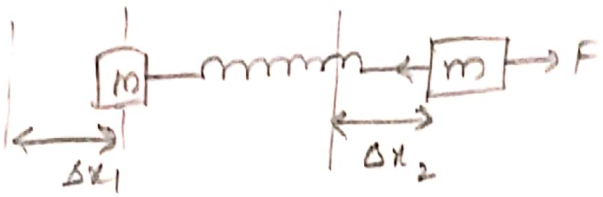
∴ position of CM at any time 't' i.e,

$$\Delta x_{cm} = \frac{1}{2} a_{cm} t^2$$

$$= \frac{1}{2} \frac{F}{m_1+m_2} (t)^2$$

$$\boxed{\Delta x_{cm} = \frac{Ft^2}{2(m_1+m_2)}}$$

②



we have,

$$\Delta x_{cm} = \frac{m_1 \Delta x_1 + m_2 \Delta x_2}{m_1+m_2}$$

$$\frac{Ft^2}{2(m_1+m_2)} = \frac{m_1 \Delta x_1 + m_2 \Delta x_2}{m_1+m_2}$$

$$\frac{Ft^2}{2m} = \Delta x_1 + \Delta x_2 \quad -①$$

$$\Delta x_2 - \Delta x_1 = x_0 \quad -②$$

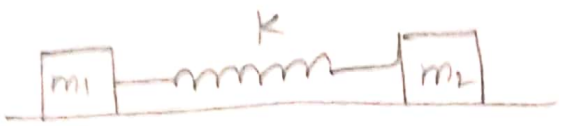
①+② $2\Delta x_2 = x_0 + \frac{Ft^2}{2m}$

$$\boxed{\Delta x_2 = \frac{1}{2}\left[x_0 + \frac{Ft^2}{2m}\right]}$$

①-② $2\Delta x_1 = \frac{Ft^2}{2m} - x_0 \Rightarrow \boxed{\Delta x_1 = \frac{1}{2}\left[\frac{Ft^2}{2m} - x_0\right]}$

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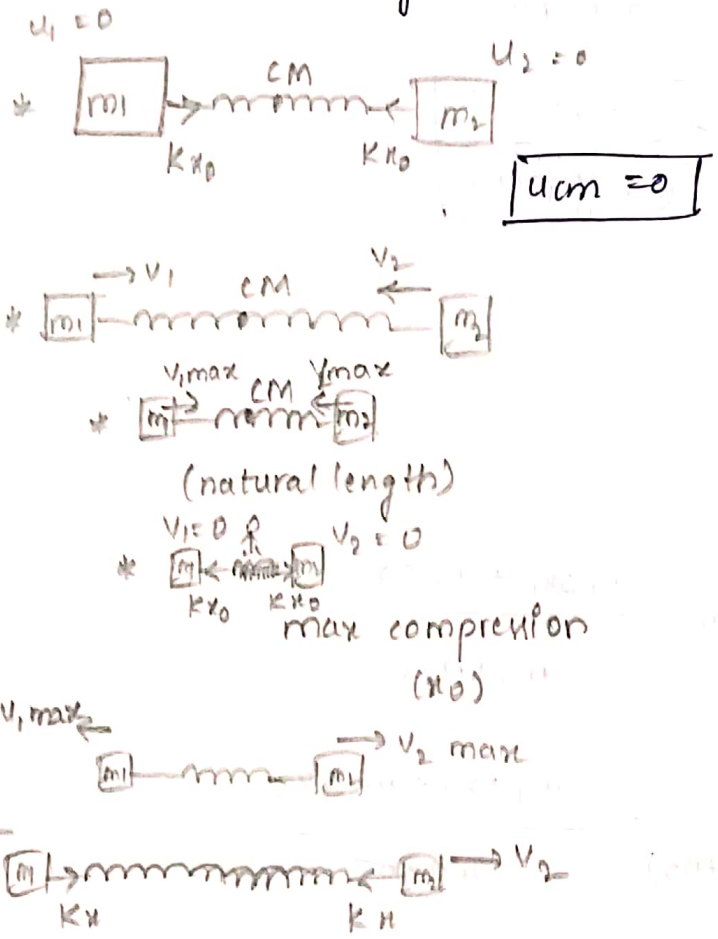
→



Given initial extension in spring = x_0

When the system is released from rest, Find distance travelled by two blocks before coming to rest.

Blocks are moving oscillating motion w.r.to cm.



Here $\Delta x_{cm} = \dfrac{m_1 \Delta x_1 + m_2 \Delta x_2}{m_1 + m_2} = 0$

$$m_1 \Delta x_1 - m_2 \Delta x_2 = 0 \quad \text{—(1)}$$

& $$\Delta x_1 + \Delta x_2 = 2x_0 \quad \text{—(2)}$$

(1) $\Rightarrow$ $m_1(2x_0 - \Delta x_2) - m_2 \Delta x_2 = 0$

$$2x_0 m_1 - \Delta x_2 (m_2 + m_1) = 0$$

$$\Delta x_2 = \frac{2x_0 m_1}{m_1 + m_2}$$

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Ily $\boxed{\delta x_1 = \frac{2m_2 x_0}{m_1+m_2}}$

→ $m_1 = 4\,kg$, $\overline{v_1} = 5\hat{\uparrow}\ m/s$, $m_2 = 2k_2$, $\overline{v_2} = 2\hat{\uparrow}\ m/s$. Find

① K.E of CM of system

② K.E of system w.r.to CM.

③ K.E of system w.r.to ground.

① $K.E_{cm} = \frac{1}{2} M |\overline{v_{cm}}|^2$

$= \frac{1}{2}(4+2)\left(\frac{4\times 5\hat{\uparrow} + 2\times 2\hat{\uparrow}}{6}\right)^2$

$= \frac{1}{2}\times 6\ \frac{(20\hat{\uparrow}+4\hat{\uparrow})^2}{36}$

$= \frac{1}{12}(20\hat{\uparrow}+4\hat{\uparrow})\cdot(20\hat{\uparrow}+4\hat{\uparrow})$

$= \frac{1}{3}(400+16) = \frac{104}{3}\,J$

② $K.E_{system/cm} = \frac{1}{2}\frac{m_1 m_2}{m_1+m_2}|\overline{v_1}-\overline{v_2}|^2$

$= \frac{1}{2}\times\frac{4\times 2}{6}\times(\sqrt{5^2+2^2})^2$

$= \frac{2}{3}(25+4) = \frac{29\times 2}{3} = \frac{58}{3}\,J$

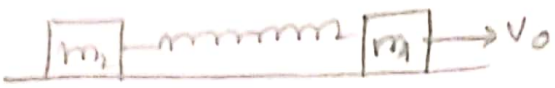
④ $K.E_T = K.E_{system/cm} + K.E_{cm}$

$= \frac{104}{3}+\frac{58}{3} = \frac{162}{3}\,J = 54\,J$

or $K.E_T = \frac{1}{2}m_1 v_1^2 + \frac{1}{2}m_2 v_2^2 = \frac{1}{2}\times 4\times 25 + \frac{1}{2}\times 2\times 4$

$= 50+4 = 54\,J.$

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* From ground frame:-



$(V_1 > V_2)$

elongation in spring = x

Here $F_{ext} = 0$

L.C.L.M, $\overline{P_i} = \overline{P_f}$

$$\boxed{m_1 V_0 = m_1 V_1 + m_2 V_2} \quad -(1)$$

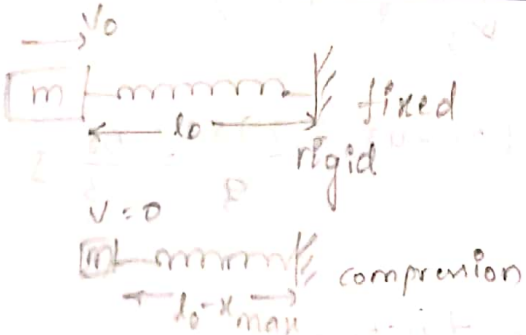
L.C.E, $$\frac{1}{2} m_1 V_0^2 = \frac{1}{2} m_1 V_1^2 + \frac{1}{2} m_2 V_2^2 + \frac{1}{2} K x^2 \quad -(2)$$

solving eq(1) & (2) we get $\boxed{V_1 \text{ & } V_2}$



L.C.L.M, $m_1 V_0 = m_1 V_1 + m_2 V_2$

$$\Rightarrow \boxed{V_{common} = \frac{m_1 V_0}{m_1 + m_2} = V_{COM}}$$



for max elongation, we use only L.C.E but not L.C.L.M bcoz one of the side is fixed.

from L.C.E, $\frac{1}{2} m V_0^2 = \frac{1}{2} K x_{max}^2$.

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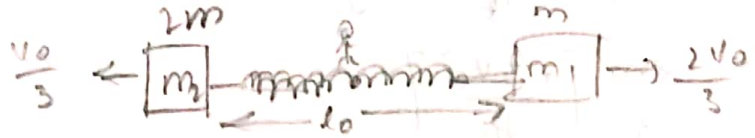
*



$$m_2 V_0 = (m_1+m_2) V$$

$$\frac{1}{2} m_2 V_0^2 = \frac{1}{2}(m_1+m_2) V^2 + \frac{1}{2} K x^2$$

* From COM frame :-



① $$V_c = \frac{m_1 V_0}{m_1+m_2} = \frac{m V_0}{3m} = \frac{V_0}{3}$$ to right.

* $$\bar{V}_{1,c} = \bar{V}_1 - \bar{V}_c$$

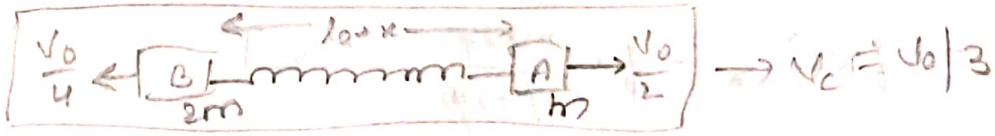
$$= V_0 - \frac{V_0}{3} = \frac{2V_0}{3}, \text{ to right}$$

& $$\bar{V}_{2,c} = \bar{V}_2 - \bar{V}_c$$

$$= 0 - \frac{V_0}{3}$$

$$= \frac{V_0}{3}, \text{ to left.}$$

*



$$\frac{2mV_0}{4} = mV_A$$

$$V_A = \frac{V_0}{2}$$

$$\bar{V}_A = \bar{V}_{A|C} + \bar{V}_C$$

$$= \frac{V_0}{2}\uparrow + \frac{V_0}{3}\uparrow = \frac{5V_0}{6}\uparrow$$

$$\bar{V}_B = \bar{V}_{B|C} + \bar{V}_C$$

$$= \frac{-V_0}{4} + \frac{V_0}{3} = \frac{V_0}{12}.$$

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* $W_{spring} = \Delta K_{system/cm}$

$$-\frac{1}{2}Kx^2 = \frac{1}{2}\frac{m \times 2m}{m+2m}\left[\left(\frac{V_0}{2}+\frac{V_0}{4}\right)^2 - V_0^2\right]$$

$$-Kx^2 = \frac{2m}{3}\left[\left(\frac{3V_0}{4}\right)^2 - V_0^2\right]$$

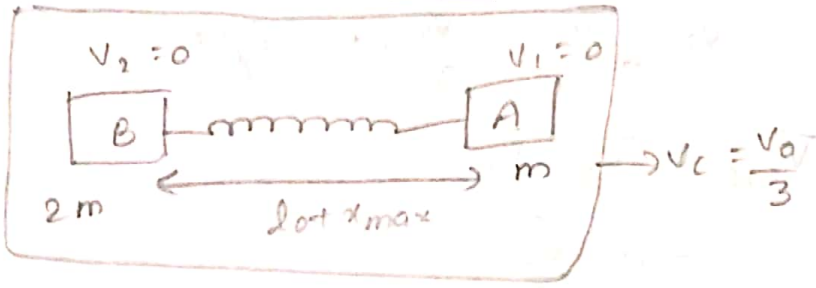
$$-Kx^2 = \frac{2m}{3}\left[\frac{9V_0^2 - 16V_0^2}{16}\right]$$

$$Kx^2 = \frac{2m}{3}\left[\frac{+7V_0^2}{16}\right]$$

$$Kx^2 = \frac{7mV_0^2}{24}$$

$$\boxed{x = \sqrt{\frac{7m}{24K}}\,V_0}$$

*



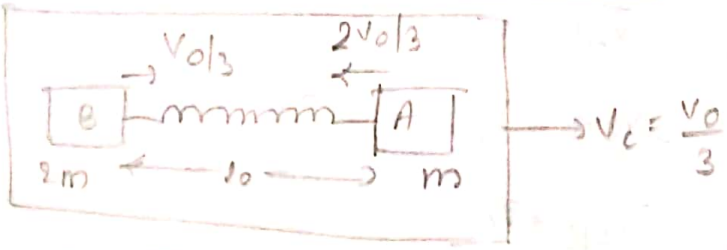
$W_{spring} = \Delta K_{sys/cm}$

$$\Rightarrow -\frac{1}{2}Kx_{max}^2 = \frac{1}{2} \times \frac{m \times 2m}{m+2m}\left[0^2 - V_0^2\right]$$

$$+Kx_{max}^2 = \frac{2m}{3}\left(+V_0^2\right)$$

$$\boxed{x_{max} = \sqrt{\frac{2m}{3K}} \cdot V_0}$$

*



$$\overline{V_A} = \overline{V_{A/c}} + \overline{V_c}$$

$$(V_A)_{max} = (\overline{V_{A/c}})_{max} + \overline{V_c}$$

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$$(\bar{V}_A)_{max} = \frac{2V_0}{3} + \frac{V_0}{3} = V_0$$

* $(\bar{V}_A)_{min} = (\bar{V}_{A/C})_{max} + \bar{V}_C$ & both are opposite.

$$= -\frac{2V_0}{3} + \frac{V_0}{3} = -\frac{V_0}{3}$$

$(\bar{V}_B)_{min} = 0$.

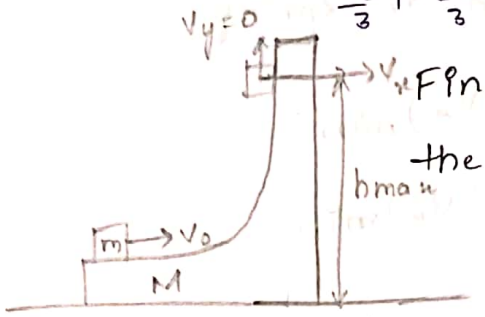
* $(\bar{V}_B)_{max} = \bar{V}_{B/C} + \bar{V}_C$

$$= \frac{V_0}{3} + \frac{V_0}{3} = \frac{2V_0}{3}.$$

$(\bar{V}_B)_{min} = (\bar{V}_{B/C})_{max} + \bar{V}_C$ & opposith

$$= -\frac{V_0}{3} + \frac{V_0}{3} = 0.$$

→



Find max height attained by the block.

* $W_g = \Delta K_{system/cm}$

$$\Rightarrow -mgh_{max} = \frac{1}{2}\left(\frac{mM}{m+M}\right)\left[0^2 - V_0^2\right]$$

$$mgh_{max} = \frac{mM}{2(m+M)} V_0^2$$

$$\boxed{h_{max} = \frac{MV_0^2}{2g(m+M)}}$$

→

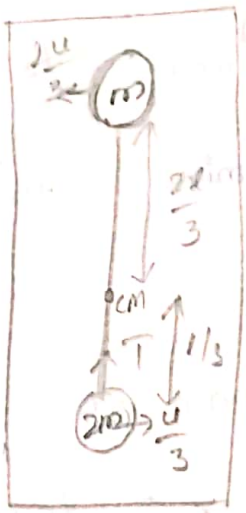


system is kept on horizontal table. Find maximum tension in the spring.

we can solve the problem by using centre of mass frame.

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$$V_c = \frac{2mu}{3m} = \frac{2u}{3}$$



Angular velocity of each mass about cm

$$\omega = \frac{\frac{u}{2}}{\frac{l}{2}} = \frac{u}{l}$$

* Tension in the string,

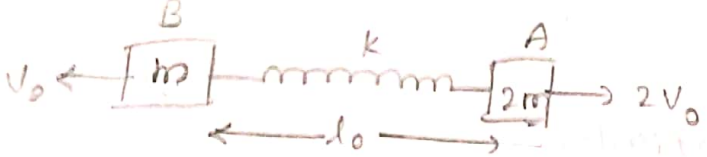
$$T = \frac{2m(l/3)\omega^2}{r}$$

$$= \frac{2mu^2}{3 \times l/2} = \frac{2mu^2}{3l}.$$

→ Find ① max elongation in spring

② $(V_A)_{max}$ & $(V_A)_{min}$

③ $(V_B)_{max}$ & $(V_B)_{min}$



$$V_c = \frac{u_m V_0 - m V_0}{3m} = V_0$$

$$\bar{V}_{1,c} = \bar{V}_1 - \bar{V}_c$$

$$= -V_0 - V_0 = -2V_0$$

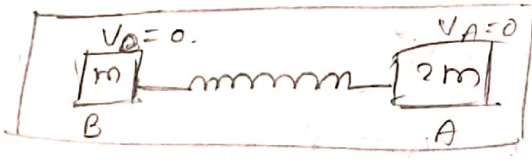
$$\bar{V}_{2,c} = \bar{V}_2 - \bar{V}_c$$

$$= 2V_0 - V_0$$

$$= V_0$$

$$\bar{V}_{1,2} = V_0 - (-2V_0)$$

$$= 3V_0$$



$W_{spring} = \Delta K_{sys/cm}$

$$-\frac{1}{2}kx_{max}^2 = \frac{1}{2} \times \frac{m \times 2m}{3m}\left[0 - (3V_0)^2\right]$$

$$\frac{1}{2}kx_{max}^2 = \frac{1}{2}\frac{2m}{3}(9V_0^2)$$

$$\boxed{x_{max} = \sqrt{\frac{6m}{k}}\,V_0}$$

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